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MACHINE LEARNING FOR SUPPORTING
SYSTEMATIC LITERATURE REVIEWS

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Dedication

I dedicate this work to my father, *Allan Kardec Afonso Borges* without him I could hardly have made this journey to a distant country and been able to study. To my mother, *Izaira Dionísia Borges*, who, although no longer present in my life today, left me an inheritance of knowledge and fine examples of how to live. I cannot fail to mention my girlfriend, whom I love dearly, *Mariana Lima Barbosa*, who had so much patience with me, helping and encouraging me through the difficult and the joyful moments of my journey. Last of all, I am grateful to God for having placed these people in my life; without Him none of this would have been possible.

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I thank the people who were part of my life in Lisbon, the city where my knowledge grew, and who helped me understand a little more about the circumstances that shape my life today and about what may yet come.

*Help the one who errs;
their feet walk the same ground, and,
if you have it in you to correct,
you have no right to condemn.*

Allan Kardec

Abstract

Due to the great demand for scientific publications, researchers find it hard to follow discoveries within the same domain. Synthesizing the investigation, either informally or through systematic reviews, it becomes increasingly resource-intensive as searches retrieve more potentially relevant information for reviewers to assess the relevance of the issue and the investigation. With the need to start medical research in the field of genetics, a small team of researchers used a series of keywords in the PubMed search engine. That returned a chain of articles, making the screening process more difficult, and not all of these texts will be relevant for the literature review, due to the quantity and in the short term it will not be possible for the reviewers to read them all in-depth. With the task of screening 17,132 articles, the reviewers use the “relevant” or “not relevant” labels to classify the information in the title and abstract fields, thus easing the burden of reading the entire content. With a very high number of texts, reviewers were able to screen 327 articles, and in this process, they obtained more studies classified as “not relevant” than “relevant”, forming an unbalanced *corpus*. Focusing on this problem a classifier is proposed that will support researchers in sorting articles, this will be trained in a supervised manner with articles that have gone through manual sorting, with a small set of training that serves to train the classifier. To solve the imbalance problem, the oversampling technique is applied, showing to be very useful when there is little information contained in the *corpus*. This resampling procedure allows a better approach to articles with text processing techniques such as text mining, enabling the use of another strategy called term frequency – inverse document frequency (TF-IDF). The methodology used for the qualification of articles is the supervised binary classification, using the support vector machine (SVM) classifier, which proved to be effective in conjunction with the probabilistic method latent Dirichlet allocation (LDA), a model for discovering thematic structures hidden in collections of texts, where the support vector machine might not be more accurate. This is intended to demonstrate that the use of machine learning methods is effective in learning by assisting investigations with systematic review.

Keywords: Machine Learning; SVM; LDA; *Corpus*; cross-validation; oversampling; summarization; literature review; machine learning.

Note on this English edition

This is an unofficial English reconstruction of the dissertation originally written and defended in Portuguese in July 2020. The layout reproduces the original as closely as possible — A4 page, Times New Roman, 1.5 line spacing, justified text, and the original chapter, figure, and table numbering — but it is a typeset reconstruction, not the original file with its text replaced. Page numbers therefore correspond only approximately to those of the Portuguese edition.

The *Abstract* above is the author’s own English text, reproduced verbatim from the original. Tables 1, 2, 4, and 5, and Figures 2, 5, and 6, were redrawn so that their contents appear in English; where a figure illustrates Portuguese morphology specifically, the original Portuguese terms are kept and an English gloss is added in parentheses, since replacing them would change what the figure demonstrates. The remaining figures are the original images: where they are screenshots of code, console output, or diagrams, any text inside the image remains in Portuguese, since altering it would mean redrawing evidence of the work as it was actually carried out. Captions are translated throughout.

The bibliography and the annex are reproduced verbatim, as they consist of citations and source code that do not change between editions. The bibliography preserves the original numbering, in which the numbers 74, 75, and 121 are not used.

Three internal cross-references that were plainly mistyped in the original have been corrected here, since leaving them would point the reader to the wrong place: in section 3.7, two references to “equation (18)” now read equation (25), the equation actually under discussion; in section 3.5.3 the reference to “equation (17)” for log perplexity now reads equation (24); and in section 4.2 a forward reference to “Fig. 28” now reads Fig. 26. Nothing else in the text was altered.

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List of Acronyms and Abbreviations

BOW	Bag of Words
IDF	Inverse Document Frequency
TF-IDF	Term Frequency – Inverse Document Frequency
LDA	Latent Dirichlet Allocation
NLTK	Natural Language Toolkit
ANN	Artificial Neural Networks
SVM	Support Vector Machines
NLP	Natural Language Processing
MeSH	Medical Subject Headings
NE	Named Entities
SR	Systematic Review

1. Introduction

A defining characteristic of modern science is that it is a cumulative achievement. The knowledge attained at any given moment rests on the vast *corpus* of knowledge previously discovered and communicated. Every researcher can echo the famous words of Isaac Newton: “If I have seen further it is by standing on the shoulders of giants” [1]. For this reason, reviews — that is, syntheses of the most relevant knowledge on a given topic — are crucial in science [2]. They make it possible to determine whether what has been published in a given field allows interpretable patterns or trends to be identified, to aggregate empirical results in a way that enables evidence-based practice (in medicine, for example), to create new theories, or to uncover questions that require further investigation [3].

There are two types of literature review [4]. The first and more common is the one carried out in any scientific article or dissertation, which, in sections titled, for example, “state of the art,” seeks to synthesize what has been published on the topic of interest, and may also provide the context of the problem to be addressed and the theoretical framework for the solution proposed to the identified problem [5]. The second type is called a systematic literature review and is itself a work of research. It seeks to synthesize a given scientific field completely and rigorously, striving to eliminate as far as possible any subjectivity in the choice of the works to be cited. A systematic literature review article aims to be a starting point for the members of a given scientific community, without any analysis of primary data taking place [6].

In fields of knowledge where time is a crucial factor, because obtaining the knowledge is urgent, systematic literature reviews take on special importance, since they place research teams at the frontier of knowledge and in possession of a reliable overview of what has been done in the past. Such is the case of the life sciences. For example, in managing the current COVID-19 pandemic, accelerated systematic reviews have been produced, notably to obtain scientific evidence regarding the use of masks [7] and regarding the effect of COVID-19 on pregnancy [8], among many others. In the same way, when health technology assessments are carried out, particularly of pharmaceuticals, conducting systematic reviews as quickly as possible is essential for patients, physicians, and industry [9].

Over the past few decades, the number of scientific articles published globally has been growing at rates of around 8% [10]. In the life sciences that growth has been somewhat higher still [10]. PubMed, the most widely used repository of articles in the life sciences (though it does not represent everything that is published), receives more than 1 million articles deposited annually [11]. Given this exponential growth in publications, it is understandable that preparing systematic literature reviews is a task that consumes a great

deal of time and resources, both human and financial, with an average duration that can range from 6 months to several years [12, 13].

The development of machine learning methods to semi-automate the preparation of systematic literature reviews has been a focus of interest in recent years, constituting a research and development subfield of its own [14], to the point that systematic literature reviews on this very subject exist, e.g., [15, 16]. Most of these techniques perform text classification, data extraction, or summarization [17]. Text classification aims to develop models that make it possible to organize documents automatically according to a predefined set of categories (for example, related to Alzheimer’s disease versus not related to that disease). Data extraction models aim to identify words, numbers, or short excerpts of text that correspond to a particular piece of desired information (for example, a chemical formula representing a protein, when the goal is to extract references to proteins, or to extract the number of people taking part in a given clinical trial). Summarization aims to synthesize a new text document whose content constitutes a summary of the original document.

The purpose of this dissertation is to present a machine learning solution to semi-automate the process of preparing a systematic literature review on the topic “Quality of life in patients with Congenital Disorders of Glycosylation (CDG) and their families.” Any systematic literature review begins by defining a set of inclusion and exclusion criteria for the articles to be considered. Part of those criteria is a set of keywords, and combinations of them, chosen so as to perform an initial extraction of articles from a repository. Naturally that extraction is very coarse, so the set of articles obtained in this way — usually numbering in the tens or hundreds of thousands — contains a high percentage of articles that are not relevant to the intended literature review.

A fundamental step in preparing a literature review is the initial screening based on the abstracts of the articles in that set. This screening represents a trade-off: assessing all of the articles, but on the basis of their abstract and title alone. In this way the number of articles can be reduced substantially with moderate effort, by limiting the analysis to the article’s abstract. It should be noted that, despite this trade-off, this is still a task that represents a great deal of effort and is normally very time-consuming. This dissertation presents a machine learning model to automate this screening on the basis of the titles and abstracts of the articles.

2. Review of the state of the art

Research challenges today call for the search for new techniques and new ideas, leading the scientific community to create the means for new technologies to emerge. As an introduction to the concepts of machine learning in systematic literature review, this chapter presents some of the reviews of related work that employ machine learning methods, together with the principles of their approach and the means for developing the topic well.

The systematic literature review aims to identify, analyze, and interpret the available evidence related to a particular research topic or phenomenon of interest, and is one of the main means of producing summarization [18], which has made this method increasingly popular, as can be seen in works such as [18, 19].

This method for discovering knowledge in texts is widely used today in fields of research such as history, medicine, biomedicine, and pharmacy [20, 21]. According to the authors in [22], the systematic review (SR) originated in the field of medicine, becoming a key fact-based research method. Investigations of this kind showed that expert opinions based solely on the experts' own medical experience were not as reliable as those grounded in the results of scientific experiments — that is, in evidence.

As the SR method developed, positive results accumulated and many specialists came to adopt it. In 2004 it was brought into line with what is practiced in medicine, with procedures being proposed for carrying out SRs in software engineering [23]. In [24] the authors show that the SR has become the best-known research method where summarization and evidence in software engineering and computer science are concerned, which has led to significant growth in its use in recent years [18, 19, 24].

Researchers and health policy makers increasingly contribute to the development of this method. As mentioned earlier, conducting an SR can be a time-consuming task, which confronts developers of clinical practice, and other decision makers, with the need to make informed decisions in a timely manner — for example, in infection outbreaks or health technology assessments in hospitals. In order to bring precision to the question of SR production, the following options were analyzed: (a) implementing process parallelization (for example, having two or more human reviewers available to screen the articles), (b) adapting and applying innovative technologies, and (c) modifying SR processes (for example, study eligibility criteria, search sources, data extraction, or quality assessment) [12].

As regards option (b), implementing SRs with the use of new technologies is a subject receiving ever greater attention, given the importance of improving it, since it helps reduce the workload and improves efficiency in production. However, according to [12],

new technologies need to be assessed for accuracy, reliability, practicality, and cost. The authors in [25] illustrate option (b) by showing that instruments such as online catalogs — the Systematic Review (SR) Toolbox site, for instance — provide a list of tools available for download in support of SRs, demonstrating the use of technological means for this type of situation. One example is the use of technological tools such as machine learning methods in areas that allow the automation of specific SR processes, particularly those involving time-consuming and resource-intensive tasks, such as language translation [12, 26], study selection, data extraction, and risk-of-bias assessment [12].

In [27] the authors propose evaluating the use of DistillerAI, a tool built into the DistillerSR literature review software [28], a commercially available web application. The authors in [27] assembled five teams of systematic reviewers who screened 2,472 proposed articles in parallel; each of these teams trained the DistillerAI tool, which replaced one human reviewer on each team and provided predictions about the relevance of the records.

The results obtained in [27] for this tool showed that its sensitivity for software-assisted screening across the five teams was inadequate to temporarily replace a human reviewer during article screening for an SR, making the tool better suited to rapid reviews, which do not require the detection of all relevant evidence, and placing semi-automation tools on a higher scale of usefulness than traditional reviews.

To address the questions raised by the authors in [27], where tools for carrying out SRs are not always adequate, new kinds of possibilities for acquiring information have been created — for example, the PubMed search engine, where researchers have easier means of finding articles grounded in biomedicine. Even so, for the SR method to be practiced well, the data gathered cannot be imbalanced, and text processing and classification methods and techniques must be used to that end.

In [29] the authors show that even experienced researchers have difficulty screening in search engines of the PubMed kind, which do not always afford the opportunity to find information that composes a balanced *corpus*. It is therefore necessary to employ new technologies, using procedures that build keyword queries, giving researchers a minimum number of articles that are relevant to the questions under investigation while generating a larger quantity of non-relevant articles, which produces data imbalance in the investigation.

In a situation like this, the use of machine learning methods can help minimize the imbalance of the data collected. This can be seen in [17, 29]. In [30] the authors devise research methods for processing the texts collected, in which they present the training of four types of binary classifiers — SVM, k-nearest neighbor, random forest, and generalized linear models with elastic net — showing combinations of these four techniques that can

solve or alleviate the class imbalance problem through text processing using text mining techniques [29].

A similarity can be observed in [30], where the researchers used the PubMed search engine to carry out screening, reducing the workload in preclinical reviews of animal studies. They applied two independent machine learning approaches to screen a large number of citations. In the first approach they used features such as the n-gram model to form the text representation sample known as Bag of Words (BoW), and the SVM classifier with stochastic gradient descent (SGD) to classify the samples. In the second approach they used the probabilistic model known as latent Dirichlet allocation (LDA) with singular value decomposition (SVD), showing the importance of words within a given article by means of the TF-IDF statistical measure. From this it can be concluded that, for carrying out the SR of articles using new technologies, there is no single way, but rather the best choice for each situation that can resolve the question of supporting the SR.

For the construction of the machine learning classifier for SR in this dissertation, the proposed solution is based on machine learning and will be strongly integrated with text mining techniques, facilitating the reduction of the workload involved in carrying out SRs. Reference [31] defines text mining as a set of processes that discovers innovative knowledge in text, which in this dissertation is accompanied by the use of NLP techniques, a computational technique involving languages [32].

Text mining techniques together with NLP assist in the process of knowledge discovery and hypothesis generation, given the volume of the literature [33]. The main objective of combining these two techniques in this work is to extract new information, such as hidden relationships in the texts between named entities, making it easier for researchers to discover systematically and efficiently, collecting, interpreting, and organizing what is needed for the investigation [34].

Reducing the screening burden draws on machine learning methods; these methods helped the computer learn on the basis of the manual screening, employing two types of label: “relevant” and “not relevant” [35]. To carry out automatic classification, the supervised learning strategy is used — a technique that can begin with a small training set and, through iteration, grow the data collection in size and usefulness, although for this work the number of articles is fixed [34, 36].

The development of supervised learning proposes the use of the SVM classifier to perform the classification of the instances. This classifier is one of the most widely used where supervised training is concerned, finding a hyperplane that separates the training instances of the not-relevant class from the relevant one by a maximum margin boundary

[37]. To refine supervised classification properly, the LDA technique is applied beforehand. The authors in [20, 39] show that the LDA technique is a probabilistic model based on a set of algorithms whose purpose is to discover thematic structures hidden in large collections of documents. The idea of hiddenness in thematic structures reflects the need to reveal terms that might go unnoticed by other techniques such as TF-IDF. LDA treats texts as a mixture of distributions of hidden topics, with each topic represented by a floating-point value. It proves promising when little manually assigned information is available, forming clusters of the information to be analyzed and increasing the performance of active learning without manual annotation.

In what follows, the development of the machine learning solution for semi-automating the SR process is explained in greater detail, helping in the search for and organization of the articles on which possible investigations into the desired topic are to be carried out, with the technologies used and the methods of using them explained more fully.

3. Problem formulation and proposed solution

The rate at which research is published is increasing extraordinarily in the health field, especially in medicine [40]. Researchers who increasingly need information quickly have difficulty keeping up with new discoveries, even within a single domain — an issue that has been emerging for several years [41]. The SR becomes ever more resource-intensive as searches retrieve a greater number of potentially important articles for reviewers to examine for relevance to the research question [42].

The solution many researchers have been turning to in order to reduce this workload in less time is based on the field of text mining, a subfield of machine learning, which facilitates carrying out SRs in a range of fields of knowledge [18, 42]. The use of text mining techniques and machine learning methods to support SRs has been proving quite effective. It is becoming an increasingly common approach for limiting the human burden and the monetary resources required, and for shortening the time needed to complete these reviews [47].

This work used information originating from PubMed, a search engine used to access citations and abstracts of research articles in biomedicine, which draws in its database on another system for searching and analyzing the medical literature, MEDLINE, whose source is the bibliographic database of the United States National Library of Medicine [43, 44].

3.1. Problem formulation

When researchers select articles to begin their studies, they encounter difficulties in screening this information, because searching in search engines using keywords returns a chain of articles many of which will be irrelevant to the SR study, making it difficult for reviewers to read them all in depth in a short space of time [45, 46]. Faced with this problem, the reviewers who built the *corpus* used in this dissertation selected a set of keywords specific to the field of preclinical studies and entered them into PubMed, which returned 17,132 articles, many of which proved to be of no interest to the research team. These articles, unsuitable for screening classification, correspond to those keywords found in fields of study from other areas connected to biomedicine. To classify the volume of articles gathered, the reviewers managed to obtain 327 articles, of which only 265 are classified as of no interest and 62 articles are classified as relevant. This shows that the choice of keywords matters if the texts that may be of interest to the researchers are to be obtained from the PubMed search engine, thereby avoiding wasted time and, possibly, imbalance in the information.

3.2. Classifier for screening articles on the basis of title and abstract

The article classification process is grounded in the title and abstract fields, using machine learning methods with the SVM classification technique. To build the *corpus* used in

this dissertation, it was necessary to develop a script in Python 3. The script accesses PubMed content through the title, abstract, and MeSH (medical subject headings) fields [52]. With a spreadsheet containing three classes of keywords (see the annex), the script forms pairs and triples of the keywords, one from each class.

For the screening process used by the reviewers, two assessment criteria were applied, “relevant” and “not relevant,” making it possible to employ the supervised classification technique.

In order to assist the classifier, text processing techniques are used, in which text mining procedures are applied together with NLP processes, making it possible to use representations of the articles as a set of words with their frequencies in the text. The BoW model is used for this representation; this model does not help in classifying the content, but it leaves a basis of statistical measures that can be used by the TF-IDF technique [20, 50, 51, 53].

With this text processing implementation in place, it becomes possible to use the LDA model, which proves promising when little assigned information is available, improving the performance of supervised learning without manual annotation, helping with the representations of the articles, and exposing texts that contain hidden information that might go unnoticed by the classifier [54].

To classify the articles with the SVM technique, the term–document matrix is required. This is a matrix of feature vectors, with word counts balanced by the TF-IDF technique and of fixed length. This matrix, which contains the classification of the articles, is a list with a small set for training the supervised classifier. With the procedures mentioned above, it is possible to use the article classifier to assist learning in support of the SR.

3.3. Text mining

Text mining is the process of discovering information in large collections of texts, and it automatically identifies interesting patterns and relationships in textual data [31]. This knowledge discovery involves a range of applications such as text analysis, information extraction, summarization, classification, clustering, and computational linguistics, among others.

Text mining uses information retrieval techniques, which involve retrieving information from stored data through user queries or preformulated user profiles [55]. Being a relatively new field of research that has recently raised a great deal of interest in industry and in the scientific communities, chiefly because of the growing quantity of information available on the internet, it has been helping to develop the SR further [20, 49, 56, 57].

3.3.1. Stages of text mining

One of the stages in the text mining process is the step for the text classification technique, which is governed by the collection of data in the form of texts through the extraction of documents that may come from the most varied sources, such as emails, text fields in databases, web pages, digitized electronic texts, and so on [49, 56, 58, 59].

There are a number of phases that are of the utmost importance to text mining processes, which may be listed as: (1) extraction, (2) text preprocessing, (3) transformation, (4) mining, and lastly (5) analysis [42, 60].

- **Extraction:** Extraction is the main phase of the text mining technique, and it forms the basis of the whole development process in this work. This segment involves the task of collecting texts, which can be done in various ways; some use third-party software such as crawlers, software that traverses internet sites with the aim of collecting data automatically. In this work, however, the mode of knowledge extraction was based on screening research in which two human reviewers were used to carry out research on articles gathered from various sources, which were then classified as “relevant” or not relevant. Once enough data has been gathered for analysis, it is possible to create a *corpus* that will serve as the basis for applying text mining techniques together with machine learning techniques [61, 62].
- **Text preprocessing:** In this phase the texts originating from extraction are processed to eliminate redundancies and filter out irrelevant information that might disturb the desired information. Types of disturbance that may be detected are: spaces, irrelevant punctuation, and special characters that confuse the algorithm during learning. This phase usually consumes a great deal of processing time during knowledge extraction, because there is no single technique that can be applied across all application domains, varying greatly with the *corpus* to be used [63].
- **Transformation:** After phases 1 and 2, this part is responsible for checking whether the textual information is sound, in the sense of whether or not the *corpus* — in which the gathered and classified articles have been brought together — needs to be modified. Often, however, some text mining and text classification work requires dimensionality reduction, or even an increase in structure, so that it is sometimes necessary to add labels — for example, creating an index for faster access, thereby better organizing the content of the *corpus* internally. That is when there is no need to balance the set of texts that make up the *corpus* on account of poor classification between the classes, which generates a tendency toward higher scores for the majority classes [57].
- **Mining:** The phase in which machine learning is applied to obtain the desired knowledge, directed to the case one intends to work on. For example, if the researcher needs to

investigate degrees of similarity in the formation of groups, then the mining algorithm chosen is clustering; if, on the other hand, the same researcher needs to examine information already formed, where the prior knowledge is based on labels, the clearest method for carrying out the mining will be classification. For this work, the classification method is used, through which the main articles belonging to the different classes are found, helping in the selection of the texts one aims to obtain more quickly [49, 63].

- **Analysis:** According to [49], analysis is the effectiveness of the result of the processes one aims to obtain after the previous phases have been applied. It is the stage at which the objective of discovering new knowledge from the database that was created — that is, the *corpus* — is evaluated. This is a cyclical process in which each stage is analyzed individually to determine whether or not it is satisfactory; if it is not satisfactory, a new cycle of processes is carried out.



Figure 1 - Illustrative diagram of the text mining methodology. Source: [61].

In Fig. 1 the life cycle of text mining processes can be seen. Since the preprocessing phase is exhaustive, the sections below are set aside, after the collection of the texts, for a more precise account of this technique, with the aim of achieving a better understanding of this part, which is so necessary where text mining for the SR is concerned.

3.3.2. Preprocessing techniques in text mining: removal of characters and spaces

This stage is considered a pre-tokenization phase, a term used to indicate the beginning of the fragmentation of a textual sequence into words or meaningful elements, called tokens [65]. The pre-tokenization stage filters out any kind of unnecessary characters that might cause noise in the processing of the *corpus*. Almost always, this stage of removing newline characters or special symbols is carried out using techniques called regular expressions, also known as regex, which constitute a language for searching, extracting, and manipulating specific string patterns in texts. It is widely used in projects involving text validation, NLP, and text mining [61, 64].

3.3.3. Tokenization

Tokenization is the process of dividing a textual sequence into words or meaningful elements, called tokens. The term token will be used a great deal in this dissertation, since at times it may carry the same sense as “word.” In fact, most of the time a token represents a word in the *corpus* [66].

The use of methods such as tokenization can give rise to certain kinds of problem, for example the use of “whitespace,” given that in certain languages the space is not used as a delimiter — in Japanese or Chinese, for instance [62]. Another problem caused by the use of tokenization is the dimensionality of the words, since dividing the text into words leads to the creation of large numbers of dimensions for analysis, though this can be corrected with dimensionality reduction techniques [63].

A simple illustration of this fact uses the following line by Fernando Pessoa: “Navegar é preciso” (“To sail is necessary”). Tokenizing the sentence yields three tokens: [Navegar], [é], [preciso]. Decomposition into tokens makes the text easier to analyze; however, according to [63] this division into tokens creates numerous difficulties, making it necessary to use other techniques such as stemming, lemmatization, and stopword removal, which are briefly described below.

3.3.4. Stopwords

Stopwords are words within the text that are of little use, supplying many tokens that have no value for the context while being useful for a general understanding of the text. Reference [57] notes that stopwords are empty words, represented by the articles, prepositions, punctuation, conjunctions, and pronouns of a specific language. Reference [67] holds that, besides not helping in the analysis for text classification, they appear in practically all of the text’s content; examples of this in Portuguese are “a,” “e,” “de,” “mas,” and “com,” among others. Removing these tokens yields a gain in performance for the SR as a whole, reducing the size of the lexicon considerably [68]. Fig. 2 shows this process, followed by stopword removal, more clearly.

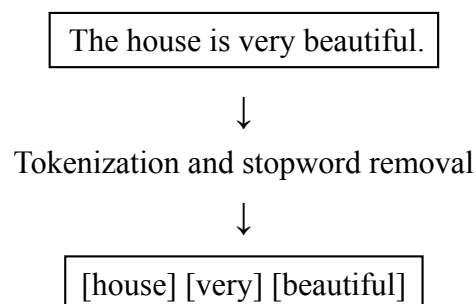


Figure 2 - Tokenization process followed by stopword removal.

Fig. 3 shows an excerpt of the classifier code containing a short list of the words chosen to be excluded in the stopwords process.

```
# NLTK Stop words
from nltk.corpus import stopwords
stop_words = stopwords.words('english')

#Essa lista de palavras básicas da língua inglesa está sendo montada para ser exclusivamente forçada a sua exclusão
#Com isso é esperado um maior acerto na classificação dos tópicos e os seus documentos. Obs: pode ser acrescentado
#mais palavras a tal lista se for necessário
stop_words.extend(['from', 'subject', 're', 'edu', 'use', 'not', 'would', 'say', 'could', 'be', 'know', 'good', 'go', 'get', 'do', 'done', 'try', 'many', 'some', 'nice', 'thank', 'think', 'see', 'rather', 'easy', 'easily', 'lot', 'lack', 'make', 'want', 'seem', 'run', 'need', 'even', 'right', 'line', 'also', 'may', 'take', 'come'])
```

Figure 3 - Example of an external list for removing English stopwords, selected in the Python 3 language.

Fig. 4 shows the internal organization of the *corpus* after it has been processed by the tokenization technique.

```
In [7]: print(data_words)

[['studies', 'in', 'the', 'modified', 'scoliosis', 'research', 'society', 'outcomes', 'instrument', 'in', 'adults', 'validation', 'reliability', 'and', 'd', 'pacity', 'study', 'design', 'observational', 'study', 'of', 'patients', 'with', 'scoliosis', 'and', 'matched', 'controls', 'objectives', 'to', 'determine', 'y', 'and', 'reliability', 'of', 'the', 'modified', 'scoliosis', 'research', 'society', 'outcomes', 'instrument', 'srs', 'for', 'use', 'in', 'the', 'assess', 'mity', 'in', 'adults', 'to', 'demonstrate', 'the', 'discriminate', 'validity', 'of', 'the', 'srs', 'in', 'differentiating', 'between', 'affected', 'and', 'ts', 'summary', 'of', 'background', 'data', 'spinal', 'deformity', 'has', 'an', 'important', 'impact', 'on', 'the', 'general', 'health', 'status', 'of', 'gnitude', 'of', 'this', 'impact', 'has', 'been', 'difficult', 'to', 'measure', 'and', 'reported', 'variably', 'in', 'the', 'literature', 'the', 'developme', 'specific', 'outcomes', 'tools', 'permits', 'improved', 'sensitivity', 'and', 'specificity', 'in', 'measuring', 'the', 'patient', 'self', 'assessment', 'status', 'an', 'instrument', 'for', 'measuring', 'disease', 'specific', 'health', 'status', 'in', 'adults', 'with', 'scoliosis', 'has', 'not', 'been', 'v', 'ls', 'and', 'methods', 'observational', 'study', 'comparing', 'the', 'health', 'status', 'of', 'adults', 'affected', 'by', 'scoliosis', 'and', 'unaffected', 'ched', 'for', 'age', 'gender', 'and', 'socioeconomic', 'status', 'pearson', 'correlation', 'analysis', 'was', 'used', 'to', 'determine', 'the', 'relations', 'domain', 'within', 'the', 'srs', 'and', 'sf', 'with', 'radiographic', 'parameters', 'including', 'sagittal', 'balance', 'coronal', 'balance', 'and', 'maj', 'ection', 'discriminate', 'validity', 'of', 'the', 'modified', 'srs', 'instrument', 'was', 'determined', 'by', 'comparison', 'of', 'means', 'between', 'aff', 'ected', 'cohorts', 'the', 'validation', 'of', 'the', 'srs', 'was', 'determined', 'by', 'criterion', 'validity', 'using', 'correlation', 'analysis', 'wit', 'domains', 'of', 'the', 'sf', 'the', 'reliability', 'of', 'the', 'srs', 'was', 'demonstrated', 'using', 'test', 'retest', 'parity', 'and', 'cronbach', 'al', 'r', 'internal', 'consistency', 'results', 'one', 'hundred', 'eighty', 'adults', 'were', 'included', 'in', 'the', 'study', 'with', 'scoliosis', 'and', 'wit', 'ith', 'scoliosis', 'scored', 'significantly', 'lower', 'than', 'unaffected', 'controls', 'on', 'every', 'domain', 'of', 'the', 'srs', 'and', 'the', 'sf', 'd', 'ceiling', 'effect', 'of', 'the', 'srs', 'were', 'less', 'than', 'observed', 'in', 'the', 'sf', 'instrument', 'pearson', 'correlation', 'analysis', 'd', 'significant', 'correlation', 'between', 'any', 'radiographic', 'process', 'measure', 'and', 'any', 'specific', 'domain', 'within', 'the', 'srs', 'or', 't', 'son', 'of', 'similar', 'domains', 'in', 'the', 'srs', 'and', 'the', 'sf', 'demonstrates', 'high', 'correlation', 'between', 'the', 'instruments', 'test',
```

Figure 4 - Sample of the tokenization of the *corpus* used in this work.

3.3.5. Normalizing the words

The technique of normalizing the words in texts may be regarded as a set of text preprocessing techniques that reduce the lexicon by identifying and grouping words that bear relationships to one another. Since there is wide variation in scope, word normalization uses a range of techniques to carry out this task, chief among them the processes of stemming and lemmatization. For this dissertation the lemmatization technique will be used, because one and the same word may have several meanings within a context. Lemmatization helps distinguish this by reducing the word to its primary root, whereas the stemming method does not use, for example, contextual information to determine the correct sense of each term, reducing the inflected (or sometimes derived) term to its trunk, base, or root — generally a written form of the word that need not carry a meaning for the context [31, 69]. Fig. 5 below exemplifies what a word looks like after the stemming process.

“Olhar” (to look) ⇒ *stemming* ⇒ “Olh”
 “Gritando” (shouting) ⇒ *stemming* ⇒ “Grit”

Figure 5 - Example of the use of the stemming method on the terms of the text.

3.3.6. Lemmatization

Lemmatization is the process of converting a word into its base form, returning it to its root and leaving it in the infinitive of the verb through the removal of suffixes, making it possible to show its semantics with a view to reducing the problem of redundant texts. The use of words in lemmatization contexts can help distinguish both unseen and ambiguous words; that is, suppose a sentence uses a word whose root is a verb, employed in some mode of verbal transition, which in certain situations causes redundancy [69, 70]. Below is an example of how lemmatization is used to reduce words to their root form without loss of the term's meaning; see Fig. 6.

“Olhando” (looking) $\Rightarrow$ lemmatization $\Rightarrow$ “Olha”
“Gritando” (shouting) $\Rightarrow$ lemmatization $\Rightarrow$ “Grita”

Figure 6 - Example of the use of the lemmatization method on the terms of the text.

3.3.7. Bag of Words (BoW)

BoW is a model that ignores word order as well as any punctuation or structural information, but retains the number of times a word appears in the text, and is represented as a term–document matrix [31]. To make the way the BoW technique works clearer, an excerpt from two texts (poems) by the author Carlos Drummond de Andrade is shown.

Text 1 (*No meio do caminho*):

No meio do caminho tinha uma pedra
tinha uma pedra no meio do caminho
tinha uma pedra
...

Text 2 (*Quadrilha*):

João amava Teresa que amava Raimundo
que amava Maria que amava Joaquim que amava Lili,
que não amava ninguém.

Using the BoW model to represent the texts shows the poems as vectors in which the number of times each word appears is counted:

Text 1: [[('caminho', 2), ('do', 2), ('meio', 2), ('no', 2), ('pedra', 3), ('tinha', 3), ('uma', 3)]]

Text 2: [[('amava', 6), ('joaquim', 1), ('joão', 1), ('lili', 1), ('maria', 1), ('ninguém', 1), ('não', 1), ('que', 5), ('raimundo', 1), ('teresa', 1)]]

Using the list indices to make it easier to classify each word of the text, it is rewritten using numeric identifiers, as shown below:

Text 1: [[(0, 2), (1, 2), (2, 2), (3, 2), (4, 3), (5, 3), (6, 3)]]

Text 2: [[(0, 6), (1, 1), (2, 1), (3, 1), (4, 1), (5, 1), (6, 1), (7, 5), (8, 1), (9, 1)]]

On the basis of the sentences, the feature vector is assembled, which is basically a list of all the words contained in both texts, without repetition:

[‘No’ ‘meio’ ‘do’ ‘caminho’, ‘tinha’, ‘uma’, ‘pedra’, ‘no’, ‘João’, ‘amava’, ‘Teresa’, ‘que’, ‘Raimundo’, ‘Maria’, ‘Joaquim’, ‘Lili’, ‘,’, ‘,’, ‘não’, ‘ninguém’, ‘.’]

This list helps reduce dimensionality through the term–document matrix process applied to the texts [71].

	w_1	w_2	w_3	$\dots$	w_n
d_1	1	0	0	$\dots$	1
d_2	0	0	1	$\dots$	1
d_3	1	1	0	$\dots$	0
$\vdots$			$\ddots$		$\vdots$
d_m	0	0	1	$\dots$	0

Figure 7 - Example of BoW. Produced by Oguri et al. Source: [71].

In the BoW model any text or sentence, document, or collection of documents is seen as a group of elements, as Fig. 7 shows; depending on the use, these elements may be simple words such as unigrams, bigrams, compound structures, and so on [31, 53].

Fig. 7 shows that d_1 represents document 1 and w_1 the occurrence of the first word in that text. The weights are represented by numbers between 1 and 0, which indicate whether or not a given word occurs in a particular text, and are used to extract information about the similarity of texts from the number of words they have in common [20]. By grounding the frequencies in weights, these aim to estimate the number of occurrences of a given word in a particular text, serving as the basis for the TF-IDF measure [58, 66, 72].

Fig. 8 shows an internal view of the matrix created by the BoW model used by the classifier proposed in this dissertation for supporting the SR, in which the frequency of the terms within the document can be seen.

	0	1	2	3	4	5	6	7	8	9 ...
0	(adult, 2)	(adult_spinal_deformity, 1)	(adults, 9)	(affected, 4)	(age, 1)	(analysis, 4)	(assessment, 4)	(background, 1)	(balance, 2)	(capacity, 1) ...
1	(age, 1)	(compared, 2)	(conclusions, 1)	(degree, 1)	(health, 4)	(matched, 1)	(methods, 1)	(observed, 1)	(one, 1)	(patient, 2) ...
2	(age, 2)	(background, 1)	(data, 3)	(determined, 1)	(eighty, 1)	(every, 1)	(greater, 1)	(hundred, 2)	(improved, 1)	(methods, 1) ...
3	(age, 1)	(disease, 1)	(domain, 1)	(included, 1)	(methods, 1)	(outcome, 1)	(population, 1)	(results, 1)	(significant, 2)	(study, 3) ...
4	(age, 2)	(analysis, 2)	(conclusions, 1)	(data, 1)	(design, 1)	(general, 3)	(high, 1)	(included, 2)	(literature, 1)	(one, 1) ...

5 rows × 394 columns

Figure 8 - Sample of the BoW to be used in the SR of this dissertation.

3.3.8. TF-IDF

The TF-IDF measure computes the importance of a word in a text on the basis of the frequency with which it appears across the whole set of documents. TF-IDF is considered the most common mode of text analysis used for SRs, which makes this feature widely used alongside machine learning algorithms [48]. This measure holds that if a word appears several times in a single document, then it is considered important and receives a high score. If, however, many documents contain that word, then it should not be used as an identifier and receives a low score [58].

To compute TF-IDF, the calculation is separated into two stages. The first defines the term frequency, TF [73, 76].

$$TF(t, d) = \frac{\text{number of times the term } (t) \text{ appears in document } (d)}{\text{total number of terms in document } (d)} \quad (1)$$

Equation 1 explains that $TF(t, d)$ corresponds to the number of times the word t appears in document d . The second stage presents the inverse document frequency part, $IDF(t)$.

$$IDF(t) = \ln \left(\frac{\text{total number of texts}}{\text{number of texts in which the term } t \text{ appears}} \right) \quad (2)$$

In equation 2 it can be seen that the inverse document frequency is given by the logarithm of the total number of documents or texts in the set divided by the number of texts in which the word appears. The score the word receives is represented as the weight of a term [77, 78].

For a better understanding of how the TF-IDF measure works, a short example is given of how to obtain the results of the weight calculations. Suppose a given document comprising 100 words in which the word “CDG” appears three times. The term frequency (TF) is given by $3 \div 100 = 0.03$. Suppose it is known that there are 10 million documents and that the word CDG appears in 1,000 of those documents. The inverse document frequency

(IDF) is computed as $\ln(10,000,000 \div 1,000) = 4$. The TF-IDF weight is then the product of these values: $0.03 \times 4 = 0.12$.

3.4. Automatic document classification

In [79, 80] the authors show that automatic document classification is a procedure that refers to the construction of classifiers through the use of inductive processes. This inductive process, called learning, builds a classifier for a class C by observing the characteristics of a set of documents previously classified under class C by a researcher in the field in question. This type of learning is called supervised, in which a new document is classified by comparison with a classifier produced and trained from labeled (classified) documents.

For the development of the classifier, two tools are used in this dissertation for this purpose: SVM and LDA. Both will be used together, with LDA serving to assist the SVM when there is little manually assigned information [42].

The input to the classification treatment is made up of several collections of data — in the case of this dissertation, texts represented algebraically by an apparatus of documents D , or *corpus*, in the following format $D = \{d_1, d_2, \dots, d_i\}$ — and the classes associated with each collection of documents in the following form, $C = \{c_1, c_2, \dots, c_k\}$ [31, 77, 122].

Classification consists in determining whether or not document d_i belongs to the class category C_k , for $i = 1, 2, \dots, n$ and $k = 1, 2, \dots, z$, these being represented as vectors $(d_1, c_1), \dots, (d_i, c_k)$, where d_i is a vector representing the terms that occur in the text and C_k is the class associated with the document, thus maintaining the following relation represented by the text preprocessing vector: $D = \{(d_1, c_1), (d_2, c_2), (d_3, c_3) \dots (d_i, c_k)\}$ [20, 81, 82].

According to [31], data classification comprises two classes, single-label and multi-label, which serve to manage the state each document is in. Information in the single-label class belongs to only one class, whereas in the multi-label case the same text or document may belong to, or be associated with, more class categories [20].

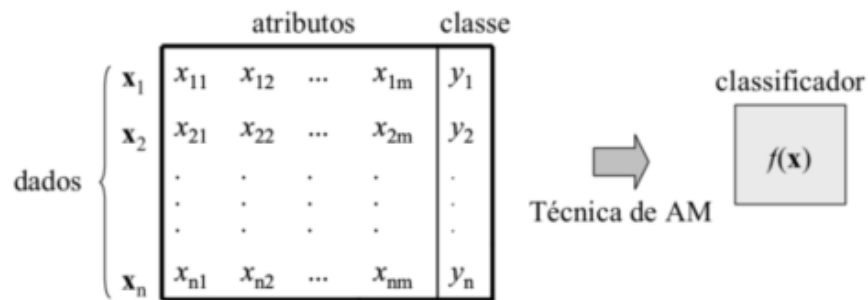


Figure 9 - Induction of a classifier in supervised learning. Source: [123].

Fig. 9 illustrates graphically what has been explained, but in this case the set D , or the *corpus*, is represented by $x_1, x_2, \dots, x_n$ — these being the documents — forming a matrix that may carry the classes, or labels, represented by $y_1, y_2, \dots, y_n$, which supply information to the classifier $f(x)$.

3.5. Classification algorithms

This chapter presents the classifier used in this dissertation and the probabilistic model employed to identify groups of topics, which will help obtain unclassified content while avoiding any kind of strong assumption about the relationship between the text and the categories, using only the distribution of words over topics.

3.5.1. SVM

SVM is a versatile classifier model, capable of linear and non-linear classification, of regression, and even of outlier detection. This model was developed by the Russian Vladimir Vapnik, and has at its foundation statistical learning theory, which is grounded in the errors of the learning technique against the validation data, or generalization error, bounded by the training error [83, 123].

The SVM technique is used in this dissertation to carry out text categorization tasks, automatically assigning one or more predefined categories to text documents. SVM can also be used in character recognition, such as signature recognition systems, in bioinformatics, assisting in the understanding of genetic structures, and in email classification, helping decide between a good email and spam, among other uses [124, 125].

The workings of SVM use the concept of supervised classification. The aim is to make the machine capable of classifying objects — or, in the case of this dissertation, texts labeled into distinct classes — from earlier examples, so that it is able to classify new inputs (new texts) into the class they will belong to. The SVM technique can be applied in two ways: use on information that is in linear form, which can be separated by a hyperplane (linear SVM), and the non-linear SVM form, where the information is in a pattern that cannot be separated by a hyperplane. This non-linear type of SVM maps the training set, which lies within its original space, into a space of higher dimensions known as the feature space [138, 140]. Fig. 10 (a) presents a training set that does not have non-linear data, and Fig. 10 (b) shows the training set in a space of higher dimensions. For this dissertation the linear SVM classifier will be used [38, 123].

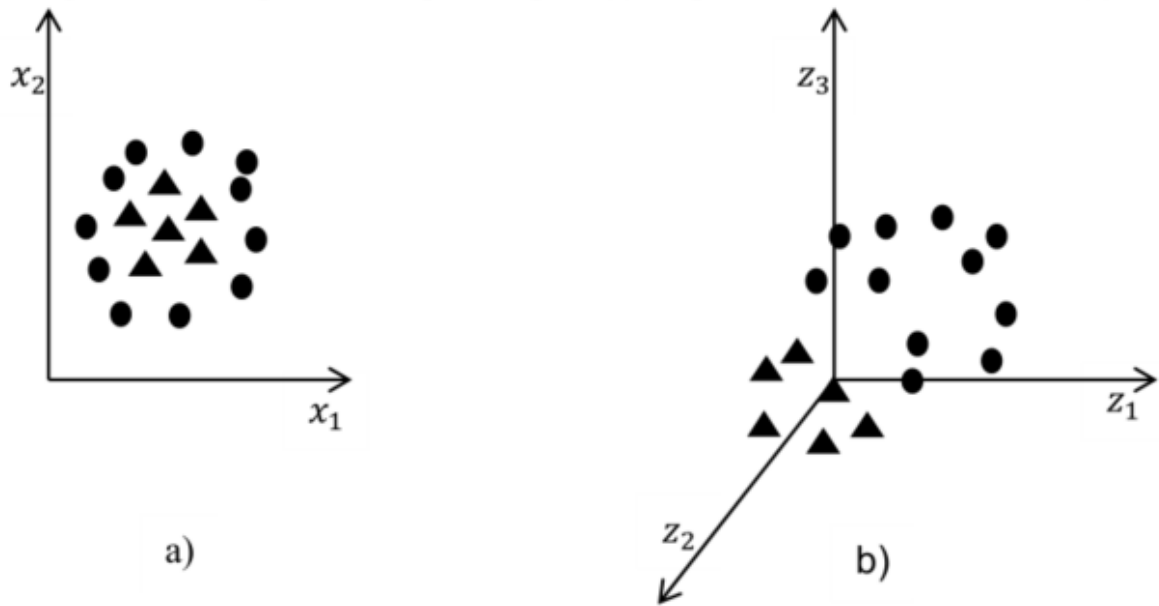


Figure 10 - Example of non-linear SVM with the feature space. Source: [139, 140].

The idea behind the SVM classifier is simple to interpret: in the training process, which contains the labeled data, the SVM classifier examines the input data and identifies patterns in two half-spaces corresponding to the two classes, with the hyperplane separating these spaces. All input data are represented as points in that space and are mapped to the output categories in such a way that the categories are divided by the largest and clearest possible gap. Fig. 11 shows what has been exemplified above [42, 48, 83, 123, 127].

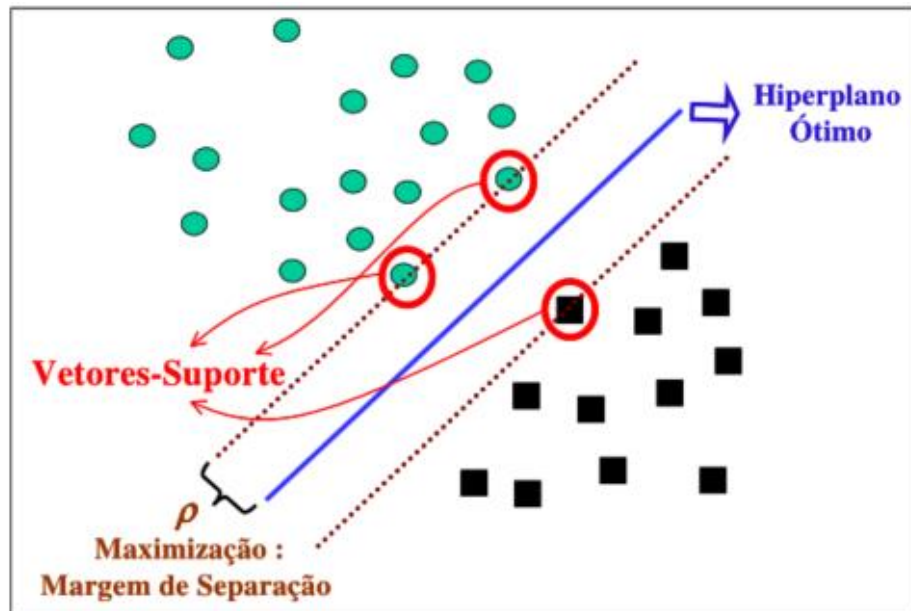


Figure 11 - Representation of the hyperplane generated by the SVM classifier and its margins.

The hyperplane is determined as a mapping function $y = f(x)$ between a multidimensional feature vector x and a quantity, or a set of them, in y . To produce the

mapping, the only information available is the training set, conceived as $D = (x_i, y_i) \in X \times Y$, $i = 1, \dots, l$, with l being the number of pairs (x_i, y_i) in the sample [38, 140].

The equation of the hyperplane in $\mathbb{R}^d$ is:

$$f(x) = wx + b = 0 \quad (3)$$

where, in equation (3):

w : is the vector representing the set of inputs, known as the weight vector;

x : are the points on the hyperplane;

b : is the distance between the hyperplane and the point of origin.

From equation (3) the hyperplane can separate the data into two regions, namely $f(x) > 0$ and $f(x) < 0$ [131, 132, 140], and the sign function can be used, as can be seen in equation (4).

$$h(x) = \text{sgn}(f(x)) = \begin{cases} -1 & \text{if } w \cdot x + b < 0 \\ +1 & \text{if } w \cdot x + b > 0 \end{cases} \quad (4)$$

Here $+1$ and -1 represent the respective classes. Given the possibility of obtaining several hyperplanes, it can be deduced that the best hyperplane for separating the classes is the one with the largest margin between the classes, which is called the optimal hyperplane. The margin is defined as twice the distance from the hyperplane (the line) to the nearest data point. In this case it can be deduced that if a margin with a large distance between the classes is obtained, then the error associated with classifying new input data will be smaller [132].

For the best hyperplane it is necessary to define the support vectors, which are the data points of both classes that lie close to the class separator (the optimal hyperplane; see Fig. 11) [133].

Fig. 11 shows the support vectors with their margins; this vector can be represented by equation (5):

$$|w \cdot x_i + b| = 1; \quad i = 1, 2, \dots, n. \quad (5)$$

where, in equation (5):

i : position of the object;

n : number of objects in the set;

x_i : corresponds to a particular data point.

This equation (5) satisfies the following constraints:

$$\begin{cases} w \cdot x_i + b \geq +1 & \text{if } y_i = +1 \\ w \cdot x_i + b \leq -1 & \text{if } y_i = -1 \end{cases} \quad i = 1, 2, \dots, n. \quad (6)$$

where:

$y_i: y_i \in \{-1, +1\}$. These values represent the classification value of x_i , defining whether the data point lies in the part greater than zero or in the part less than zero of the classifier. Equation (6) can be summarized in equation (7):

$$y_i(w \cdot x_i + b) - 1 \geq 0; \quad i = 1, 2, \dots, n. \quad (7)$$

Referring to the constraints $y_i(w \cdot x_i + b) - 1 \geq 0$:

$$y_i(w \cdot x_i + b) - 1 \geq 0, \quad \forall i = 1, \dots, n. \quad (8)$$

The optimal hyperplane is found by maximizing the margin. This maximization of the margin is obtained by minimizing, or reducing, $\|w\|$; see equation (9) [131]:

$$\text{Minimize}_{w,b} \frac{1}{2} \|w\|^2 \quad (9)$$

where:

$\|w\|$: is the Euclidean norm of w , $\|w\| = \sqrt{w * w}$.

In this way it can be guaranteed that no data point will lie between the margins; this is the characterization of linear SVM with hard margins. In order to solve the problem of minimizing $\|w\|$ subject to the constraints of equation (9), the method of Lagrange multipliers [134] is used, which supplies the maximum and minimum of a function subject to a constraint.

$$L(w, b, \lambda) = \frac{1}{2} \|w\|^2 - \sum_i \lambda_i (y_i (w \cdot x_i + b) - 1) \quad (10)$$

where:

λ_i : are the Lagrange multipliers.

For an optimal solution, it is necessary to minimize the Lagrange function by maximizing the Lagrange multipliers (λ) and minimizing w and b [135]; the following optimization problem was obtained:

$$\text{Maximize}_{\lambda_i} \sum_{i=1}^n \lambda_i - \frac{1}{2} (w \times x_i + b) \quad (11)$$

$$\text{With the constraints: } \begin{cases} \lambda_i \geq 0, \forall i = 1 \\ \dots \\ n \\ b = 0 \end{cases} \quad (12)$$

$$w = \sum_{i=1}^n \lambda_i x_i y_i \quad \text{and} \quad b = \sum_{i=1}^n \lambda_i y_i \quad (13)$$

The optimization problem seen in equation (13) has a single global maximum [137] that can be found, and may be expressed in equation (14):

$$h(x) = \text{sgn} \left(\sum \lambda_i y_i (x_i x) + b \right) \quad (14)$$

With the function λ_i different from zero, these are the data points called support vectors, being the points closest to the hyperplane; for the remaining data points $\lambda_i = 0$ [136]. The classifying function is therefore:

$$f(x) = \sum \lambda_i y_i x_i x + b \quad (15)$$

This confirms that the hyperplane obtained by the equations explained above is able to classify the data that fall within the parameter of equation (3), demonstrating that these data are linearly separable [137, 140].

3.5.2. LDA

Latent Dirichlet allocation, better known as LDA, is a probabilistic topic model used for NLP processing and content-based topic extraction, which links the category of unclassified text learning. Topic modeling is represented as a distribution over words, and its basis is tied to the Naive Bayes classifier, itself a probabilistic classifier. This technique (LDA) arose from the need to extract information from texts around the year 2003, in order to carry out topic modeling on the basis of probability, and is today one of the most popular techniques [39, 40, 54].

To identify topics, the LDA model uses clustering methods to identify meaningful groups of articles. In [39] the authors show that the topic model appears only during processing (which is why it is called latent). LDA is not a classifier in itself, but it uses a topic generation approach and is widely used to avoid any strong assumption about the relationship between the text and the categories, using the distributions of words over topics as a mathematical model. LDA does not need to have known class labels, as in some types of supervised classifier where pattern deductions are made. This whole process is described through probabilistic and statistical models in order to identify groups of topics.

The exploration of large volumes of data with little assigned information in topic discovery is demonstrated in [39, 42], where structures with semantic values — which in the context of text mining form groups of terms that frequently occur together — when analyzed give indications of a theme or subject, occurring in subsets of documents. The term topic is defined as a set of words that bear relationships within the document or text from which they are automatically extracted.

In order to understand better how the probabilistic model works, suppose that a *corpus* containing reviews of articles has been supplied, comprising several texts. The review texts submitted by reviewers over time contain many terms, some of which are used across several topics. A topic identified by the LDA probabilistic method may represent reviews for an

individual article, or it may represent a group of article reviews. For LDA, the topic itself is merely a probability distribution over time for a set of terms [84, 85].

Terms are rarely exclusive to any one text; they may refer to other texts or be general terms that apply to everything — examples would be “good,” “wonderful,” “happy,” “sadness.” Other terms may be corrupted or meaningless words, or words with unnecessary characters, which are referred to as noise. It is important to understand, however, that the LDA method is not intended to capture every word contained in the *corpus*, nor to understand how the words are related beyond their co-occurrence probabilities. It can only group words that have been used in the target domain.

After the term indices have been computed, individual lines of text are compared using a distance-based similarity measure to determine whether two parts of the text are similar. For example, one may find that the article has several names that are strongly associated, or one may find that strongly negative terms are generally associated with a particular article. The similarity measure can be used to identify related terms and to create recommendations [39, 40].

Mathematically, LDA modeling has at its basis the Dirichlet distribution, denoted $\text{Dir}(z, \alpha)$, and according to [18]:

$$\text{Dir}(z, \alpha) = \frac{1}{B(\alpha)} \prod_{k=1}^K z_k^{\alpha_k - 1} \quad (16)$$

Here $z = (z_1, z_2, \dots, z_K)$ is a K -dimensional variable, with values $0 \leq z \leq 1$ in $[0, 1]$ and $\sum_{i=1}^K z_i = 1$. In this context $\alpha = (\alpha_1, \dots, \alpha_K)$ are the hyperparameters of the distribution. The function $B(\alpha)$, called the Beta function, can be expressed in terms of the gamma function Γ :

$$B(\alpha) = \frac{\prod_{\kappa=1}^K \Gamma(\alpha_{\kappa})}{\Gamma(\sum_{\kappa=1}^K \alpha_{\kappa})} \quad (17)$$

LDA does not use the semantics used by humans; these are accessed by the algorithms through BoW techniques, by which word order is ignored [50].

The sampling of the topic distribution is carried out by the Dirichlet distribution, which is the multivariate generalization of the Beta distribution, used to allocate the words of different topics that will populate the documents. This gives an indication that the intent of this algorithmic model is to allocate the latent topics that are distributed according to the Dirichlet distribution, which in turn is a multivariate discrete distribution [39, 86].

Fig. 12 represents the Bayesian network graphically. In this network, each vertex corresponds to a variable and each edge to a dependency relation. In this representation,

instead of showing each variable repeatedly, a rectangle is used to group variables into a subgraph that repeats. The number of repetitions is labeled at the bottom of each rectangle.

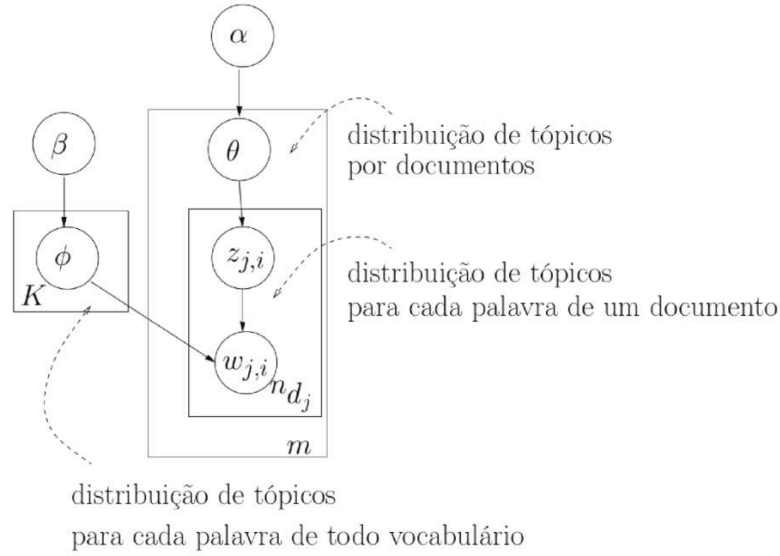


Figure 12 - Graphical model of the Dirichlet distribution. Adapted from source: [39].

LDA contains three hierarchical levels and is strongly influenced by the Bayesian model, as can be seen in Fig. 12. In the first stage there is a representation of the topic distribution across the whole collection of documents; the second stage of levels includes the distribution of topics for each document. At the last level, the distribution of topics is repeated internally for the words in a document. With this last level it becomes possible to represent a document as a mixture of topics [129].

Fig. 12 can be better described through the following key to its variables [39]:

- n — number of words in the vocabulary.
- m — number of documents.
- K — number of topics.
- n_{d_j} — number of words in a document d_j , where $1 \leq j \leq m$.
- θ — distribution of topics per document.
- φ — distribution of topics over the words of the vocabulary.
- θ_j — vector with the proportion of topics for document d_j , where $1 \leq j \leq m$.
- φ_k — vector with the proportion of vocabulary words for topic k , where $1 \leq k \leq K$.
- α — prior of the Dirichlet distribution, related to the document–term distribution.
- β — prior of the Dirichlet distribution, related to the topic–word distribution.
- w_i — i -th word of the vocabulary, where $1 \leq i \leq n$.
- $w_{j,i}$ — word w_i observed in document d_j , where $1 \leq j \leq m$ and $1 \leq i \leq n$.
- $z_{j,i}$ — topic distribution associated with the word $w_{j,i}$ in document d_j , where $1 \leq j \leq m$ and $1 \leq i \leq n$.

Looking at Fig. 12, it can be seen that the hyperparameters represented by α and β influence the level of collections of documents. If, in the LDA modeling, the value of α is

high, it means that each document will probably contain a greater mixture of topics; with the same parameter, if α is low, the LDA model will eventually have a mixture of few topics, resulting in greater concentration in those topics. A high value of the parameter β makes it likely that each topic contains a mixture of many words; conversely, a low value of β indicates topics formed with few words, yielding a weak topic model [39, 129]. In later chapters the implementation of the LDA model through the Gensim library will be presented, together with the bubble charts that the library makes available.

3.5.3. Performance measures for the classification algorithm

For the algorithm supporting the SR to work well, a number of methods are used to give greater accuracy in classification, validating whether the classifier is performing its task correctly. To classify the texts as “relevant” or “not relevant,” it is necessary to use specific measures and techniques such as cross-validation in order to obtain good results. Where supervised classification is concerned, it is necessary to use measures called metrics, which check whether the classifier is able to characterize new examples when they are presented to it.

In most cases the evaluation metric uses a matrix when supervised, binary classification is involved, with correctly and incorrectly classified samples. This is called the confusion matrix, a table frequently used to describe the performance of a classification model on a dataset [38, 86].

The general idea of the confusion matrix is to ascertain the number of times the “relevant” class is classified as “not relevant”; to reach that conclusion, the confusion matrix uses a set of predictions so that they can be compared with the actual values. This set of predictions can be described as:

Correct class	Positive	Negative
Positive	True Positives (TP)	False Negatives (FN)
Negative	False Positives (FP)	True Negatives (TN)

Table 1 - Metrics used for supervised classification.

Described in more detail below:

- TP (True Positive) = number of articles of the “relevant” class classified as articles of the “relevant” class.
- FN (False Negative) = number of articles of the “relevant” class classified as articles of the “not relevant” class.
- TN (True Negative) = number of articles of the “not relevant” class classified as articles of the “not relevant” class.

- FP (False Positive) = number of articles of the “not relevant” class classified as articles of the “relevant” class.

		<i>Predicted values</i>	
		Relevant	Not relevant
<i>Actual values</i>	Relevant	TP	FP
	Not relevant	FN	TN

Table 2 - Confusion matrix showing the categories.

from which the explanation of the following metrics can be seen more precisely:

$$FP = N - TP \quad (18)$$

where the number N is the number of “relevant,” of the “relevant” class type.

$$FN = M - TN \quad (19)$$

where the number M is the number of “not relevant,” of the “relevant” class type [54]. The confusion matrix provides a great deal of information that can give rise to misunderstanding when the results are explored; for this type of problem the issue can be resolved by using metrics, such as the classifier’s *accuracy*, *precision*, *recall*, and *F1 score*, explained as the ratio between the number of positive examples correctly classified and the total number of examples identified as positive by the classifier [38, 54, 86].

$$\text{precision} = \frac{TP}{TP + FP} \quad (20)$$

Precision yields the percentage of articles that were classified by the model as belonging to the “relevant” class, giving weight to information of the positive class (TP); the higher the precision, therefore, the less effort will be required to analyze the articles of the “relevant” class [38].

The disadvantage of this measure is that it does not take into account the articles of the “not relevant” class. According to the authors of [84, 86], the perfect way to achieve precision would be to make a single positive prediction and ensure that it is correct (precision = $1/1 = 100\%$), which is not very useful for the investigation, leading the classifier to ignore all the “not relevant” parts and less of the “relevant” ones. From this point of view, precision is used together with another metric called recall, which also goes by the designations *sensitivity* or *true positive rate* (TPR), representing the proportion of true positives that were correctly identified.

$$\text{recall or TPR} = \frac{TP}{TP + FN} \quad (21)$$

For the classifier's percentage of correct calls, accuracy is used, which considers the number of correct calls over the positive and negative samples. This measure addresses the whole objective of the classification work, thereby coming closer to the human capacity to assess the subjectivity of a text correctly [54, 87, 88].

$$\text{accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (22)$$

In some circumstances it is necessary to have a metric that indicates the overall quality of the supervised classification model and that works well even with datasets that have disproportionate classes. This metric combines precision and recall and is called F1. It is a harmonic mean of precision and recall, whereas the regular mean treats all values equally. The harmonic mean gives more weight to the lower values, resulting in high F1 scores if recall and precision are both high as well. In short, according to the author of [38], F1 is a better metric than accuracy chiefly in cases where false positives and false negatives have different impacts on the model, since F1 creates a result out of those divergences.

$$\text{F1} = \frac{2 * \text{precision} * \text{recall}}{\text{precision} + \text{recall}} \quad (23)$$

Besides the supervised classification metrics, this dissertation also uses evaluation measures for the probabilistic model, such as the model's log perplexity and topic coherence. These measures provide a convenient way of judging how good a particular topic model is.

Log perplexity measures the constraints expressed by the model. From the point of view of the recognition task, it can be said that the language model reduces the number of word options during the recognition process, so that it may be interpreted as the average number of word options. Perplexity measures the difficulty of the recognition task: the lower the log perplexity, the lower the error rate [89].

$$2^{H(p)} = 2^{-\sum_x p(x) \log_2 p(x)} \quad (24)$$

where:

- $H(p)$: is the entropy (in bits) of the distribution;
- p : discrete probability;
- x : is a variable over the events.

Note: entropy, in this case of log perplexity, belongs to the field of information theory, and is defined as a way of determining the average degree of uncertainty with respect to the information sources [90].

Topic coherence is a measurement of the quality of a topic's property that allows sense to be made of it, establishing a relation between its parts and between the text itself to which it belongs and the situation of its occurrence. In another context, coherence can be

seen as the measure of the correlation between the phases of each topic at different points of a text. Topic coherence can be divided into the following segments [91, 92]:

- **Segmentation:** where the topics are divided into several parameters, assuming that the qualities of the topics in each parameter are different.
- **Probability estimation:** where the quantity of topics in each parameter is measured.
- **Confirmation measure:** where the quality of the topic (according to a given metric) in each parameter is measured and a number is assigned.
- **Aggregation:** the gauge by which these quality numbers are combined in a certain way (arithmetic mean) to create a single number.

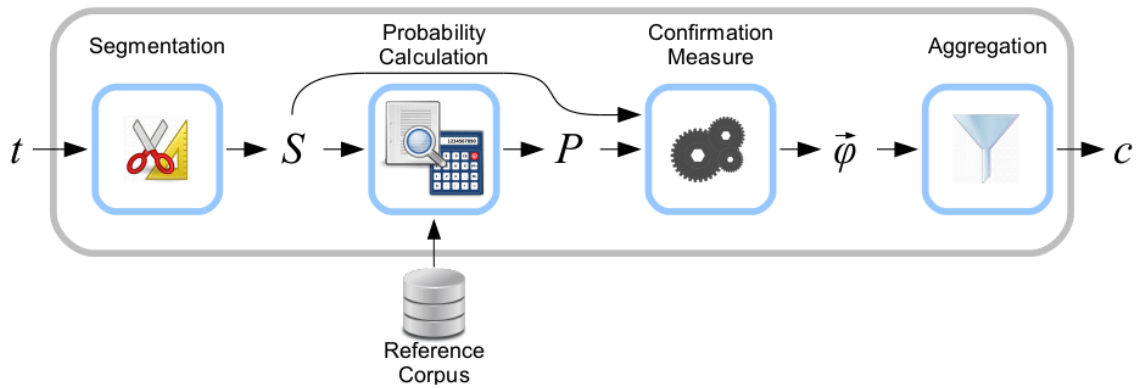


Figure 13 - Function in which the development of topic coherence can be observed. Source: [91].

This can be described according to Fig. 13 as:

- t : topics coming from the topic model;
- S : segmented topics;
- P : computed probabilities;
- $\vec{\phi}$ (phi vector): a vector of the “confirmed measures” coming out of the confirmation module;
- C : the final coherence value.

3.6. Balancing technique

To balance the information contained in the dataset, various balancing techniques may be used, the most common being random undersampling and random oversampling. For the purposes of this dissertation the random oversampling technique was adopted, as better suited to the nature of the chosen *corpus*.

The random undersampling procedure is a technique that randomly reduces the majority samples and equalizes them with the quantity of the minority ones. The idea is that the data of the dominant class contain redundant records, and that a smaller, balanced quantity of data benefits the performance of the model and makes it easier to explore. The disadvantage is that information important to the majority class may be lost [54].

The random oversampling technique, by contrast, is the opposite of random undersampling. Random oversampling randomly replicates the minority class until it equals the majority class, or until it is sufficient for the data to be balanced. This choice is based on the set of information in the *corpus*, which is a relatively small set, with rare classes containing few articles.

a few dozen records (articles). See Fig. 14 for a visual illustration of these two techniques [54].



Figure 14 - Example of the oversampling and undersampling techniques.

The random oversampling resampling technique takes into account the weight (p) of each sample of the minority class, with $p = 1$, given by the resampling algorithm that will appear in the loss function. The loss function is the optimization of the classification and regression criterion used by the oversampling resampling algorithm [54]; it weights the function in such a way as to discount errors for records with low weights in favor of records with higher weights. Many classification algorithms assume a weights argument that allows records to be weighted up or down. The weight (p) for the majority class is defined as $1/p$, so that the minority class is increased to match the proportion of the majority class; in the end the total sum of the weights is practically the same [54, 93, 94].

The advantage of using random oversampling resampling is that no information is lost, but there may be fitting issues, known as overfitting of the data, since this multiplication or replication proves ineffective for predicting new results [94].

3.7. Summarization of articles

The process of transcribing a text into a shorter version is called text summarization, and it can be of two types: extractive and abstractive. In the first, sets of important sentences from a document are produced without alteration; in the second, there is a process of improving the coherence of the text, eliminating redundancies and making the sentences more concise, which comes to create new clauses [96].

In the summarization process, reducing the quantity of text in a document while preserving its key meanings is immensely useful for trying to discover whether or not a long document meets the reader's needs and whether it is worth reading the rest of that information. With long texts, the summarization algorithm processes and summarizes the document in the time the reader would take to read the first paragraph. The main key point is to reduce the size and the level of detail of a document while preserving the main points and the overall meaning [96, 97].

This dissertation uses summarization of the extractive type, with the abstract as its key point of summary within the articles. This course was adopted for the following reason: since the abstract is a part of the article that already presents a summary of what is to be shown, summarizing on top of this part works well for articles with a long abstract, which in many circumstances confuses the reader with the article's introduction.

With a view to easing the human reviewers' screening work by reducing the reading burden, summarization based on this aspect was investigated in this dissertation, using for this purpose the Gensim library, which has at its foundation rows of text sentences and applies a variation of the TextRank algorithm [98].

Recognition of the entities of each summarization of the abstracts is also employed, using the spaCy library for this purpose together with its displacy function. Reference [99] shows that Named Entities (NE) are represented by one or more rigid designators in a given text. Some of the most common NE types are: proper nouns, such as names of people, organizations, and local entities; temporal entities such as dates, time, day, year, and month; and numeric entities, such as measurements, percentages, and monetary values [99].

The TextRank method is an unsupervised, general-purpose algorithm in the field of summarization, designed for specific NLP tasks. The main idea is the ranking of text based on graphs, its concept being grounded in mathematics, in a subfield of graph theory [100, 101, 102, 103].

The TextRank algorithm has at its basis the PageRank algorithm, another algorithm used to carry out website ranking searches, widely employed by the Google search engine. This algorithm extracts key phrases by exploring the structure of the text in question, in order to determine key phrases that appear essential to the text, in the same way as PageRank [104].

In order to understand the TextRank algorithm better, a short account of its structure in mathematical terms is required [105]:

$$C(N_i) = (1 - d) + d \sum_{N_j \in E_n(N_i)} \frac{p_{ji}}{\sum_{N_k \in S_a(N_j)} p_{jk}} C(N_j) \quad (25)$$

According to some notations, the following can be observed:

- $C(N_i)$: ranking of the nodes;
- N_i : node;
- d : d is a damping factor intended to include in the model the probability of a random jump from one node of the graph to any other;
- N_j : nodes belonging to the set $E_n(N_i)$;
- N_k : nodes belonging to the set $S_a(N_j)$; these are the total nodes that each node N_j recommends;
- p_{jk} : weight of the recommendations.

As explained in [80], equation (25) represents a graph model that can be defined as $G = (N, A)$, these being the parameters of a directed and weighted graph — that is, it assigns weights to the recommendations of each set of nodes. N is the set of nodes and A the edges, where A is a subset of $N \times N$. The node N_i is represented by the set $E_n(N_i)$, which point to it, making a self-reference.

$C(N_j)$ is the set of rankings of each of the nodes N_j belonging to the set $E_n(N_i)$ — that is, the nodes that recommend N_i — with p_{ji} as the weights aggregated to the indications. N_k represents the nodes belonging to the set $S_a(N_j)$; these are the total nodes that each node N_j recommends, the weights of those recommendations being represented by p_{jk} [100].

In equation (25) an important factor to note is the variable d , an item that helps damp the model on the probabilistic side when there is a random jump from one vertex of the graph to any other. In a usual situation while browsing the internet, this factor represents the probability of the user reaching a page by means of a link located on the current page, with $(1 - d)$ being the probability of the user jumping to the next page at random, unlinked to the page of origin. According to the TextRank documentation, the original definition of the factor d is 0.85, lying between the bounds of 0 and 1.

In this way, the TextRank algorithm is independent of training, which is required in supervised classification. It can be run on any random part of the text, determining the emergence of intrinsic texts derived from the simple basis of the properties of the text to be analyzed, which makes it a highly adaptable algorithm for use with new domains and languages [104].

4. Presentation and discussion of results

This chapter presents the development of the classifier for machine learning in systematic literature review, taking a technical approach to the use of the programming methodologies, as well as the details of how the *corpus* is constituted and the difficulties of classifying on top of it, together with the use of the metrics for the subject under discussion.

4.1. Organization and balancing of the corpus

The *corpus* of this work is made up of a collection of articles based on the searches carried out against the database of the PubMed search engine, forming a list in the Python 3 language. Four positions are found in the *corpus*, subdivided into: the index at position zero, the classification of the article at position one, the title at position two, and lastly the abstract at position three.

Index [0]	Classification [1]	Title [2]	Abstract [3]
-----------	--------------------	-----------	--------------

Table 3 - Organization of the corpus fields in the list.

Considering the classification methodology in which the articles were assessed in the “relevant” and “not relevant” categories, the set of classified articles, as already noted in section 3.1, is not balanced, containing more “not relevant” classifications than “relevant” ones, and thereby causing an imbalance in the classification results. To balance this *corpus*, some adjustments to its structure were necessary, such as converting the information base from a list format into a format more tractable for the classifier — the dataframe format of the pandas library, for example — before it passes through the text processing carried out by the text mining techniques. This makes the information more accessible to the procedures used by the classifier.

The *corpus* used in this dissertation is made up of two files. One contains the articles selected and classified by the screening team, stored in a spreadsheet file, as can be seen in Fig. 15, with a caption describing the “.xlsx file containing classified articles.” The second file contains assorted articles from the field of biomedicine but without classification, shown in Fig. 15 with a caption describing the “.dat file containing unclassified articles”; this file originates from the script developed in Python 3 to access the PubMed database, as described in section 3.2.

In [10]: records		Ficheiro .dat contendo artigos não classificados	
Out[10]: [{"PMID": "21372733", 'OMI': 'HLJ', 'STAT': 'HEBLINE', 'DCOM': '20120224', 'LR': '20111018', 'IS': '1550-5049 (Electronic) 0889-4655 (Linking)', 'VI': '26', 'DP': '6', 'DP': '2011 Nov-Dec', 'TI': 'Prevalence of obstructive sleep apnea, associated risk factors, and quality of life among Indian congestive heart failure patients: a cross-sectional survey.', 'PS': '452-9', 'LID': '10.1097/JCN.0b013e31820a040e [doi]', 'AB': 'BACKGROUND AND OBJECTIVES: Stable congestive heart failure (CHF) patients frequently suffer from sleep apnea, which worsens the prognosis of heart failure and affects sleep quality of the patient. The present study was undertaken to assess the prevalence of obstructive sleep apnea (OSA) and associated quality of life (QoL) among CHF patients and to ascertain the relationship of OSA with excessive daytime sleepiness and selected demographic, clinical, and anthropometric characteristics. SUBJECTS AND METHODS: Fifty CHF patients and 50 healthy controls were conveniently recruited at CHF clinic, cardiothoracic and neurosciences center outpatient department, All India Institute of Medical Sciences, New Delhi, India. Participants were in		Ficheiro .xlsx contendo artigos classificados	
File name	Decision	Decision is one of the following: 0 -> do not select to full read, 1 -> select to full read	
		Carlota	
138	1	138	1 Include
192	0 not adequate for PMM2-CDG	192	0 Exclude - about cervical cancer and we didn't include cervical issues in our list of symp
197	0 general questionnaire EQ-5D	197	0 Exclude - EQ-5D
198	1	198	1 Include - https://baynav.bopdhh.govt.nz/media/2267/osa-18-score-sheet-paediatric-
229	0 cost effectiveness analysis	229	0 Exclude
231	1 couldn't access the questionnaire but many CF patients have digestive problems so it might be	231	0 Exclude
406	0 it does not state which questionnaire it is so probably is not validated	406	0 it does not state which questionnaire it is so probably is not validated
415	0 clinical questionnaire	415	0 Exclude
432	0 not in english	432	0 Exclude - not in english
545	0 mental health	545	0 Exclude - depression was not one of the symptoms
554	1 this questionnaire score negatively correlated with QoL so it is interesting, http://www.scielo.	554	1 Include
562	0 not in english	562	0 Exclude - not in english
578	0 not in english	578	0 Exclude - not in english
		Rita	
138	1	138	1 Include
192	0	192	0 EORT
197	0	197	0 Exclu
198	1	198	1 The C
229	0	229	0 Exclu
231	0	231	0 Exclu
406	0	406	0 Epide
415	0	415	0 Exclu
432	0	432	0 Exclu
545	0	545	0
554	1	554	1 Includ
562	0	562	0 Not it
578	0	578	0 Not it

Figure 15 - View of the files that make up the corpus.

The file containing the classified articles is opened and processed so that its content can be extracted, in order to check whether these data contain any kind of junk, such as null values or even the question-mark characters used by the reviewers to indicate that there is indecision about the classification of a particular article. Once this process is complete, two lists are created: the first containing the classification and the second listing the indices of each article, making it possible to organize the content by pairing the elements contained in both lists, as can be seen in Fig. 16.

```
#####Leia os dados de classificação (os conjuntos de treinamento, validação e teste) #####
os.chdir(os.path.dirname(patchExcel))
df = pd.read_excel('ArticleSelection_Sample_CP_RF_SB.xlsx', 'Indexes', index_col=None, header=None)

df2 = df[df[1].notnull()] # Busca se há valores nulos em alguma coluna
isnotqm = df2[1]!='?' # Busca se há caracteres de '?' utilizado para informar indecisão na classificação
df3 = df2[isnotqm]
listinds = df3[0].tolist() # Definindo Lista para conter valores de índice
listclasses = df3[9].tolist() # Definindo Lista para conter valores de classificação

del listinds[0]
del listclasses[0]
indsAndClasses = zip(listinds,listclasses) # Juntando os campos contendo os índices de cada artigo e sua classificação
```

Figure 16 - The .xlsx file being processed to remove irrelevant content.

Continuing with the development, the second file, containing the unclassified articles, is opened. It then becomes possible to merge the two files into a single list, with the fields selected through an algorithmic structure so as to separate the classified articles from the unclassified ones into a new list containing index, title, and abstract, as shown in Fig. 17.

```

os.chdir(os.path.dirname(patchDAT))
with open("records.dat", "rb") as fp: # Unpickling
    records = pickle.load(fp)

"""Formando o corpus contendo
os artigos classificados"""
data = [[ic[0],ic[1],records[ic[0]]['TI'],records[ic[0]]['AB']] for ic in indsAndClasses]

```

Figure 17 - Loop structure for organizing the new Python list (corpus).

```

[[138, índice dado pelos revisores
1, Classificação do artigo
'Studies in the modified Scoliosis Research Society Outcomes Instrument in adults: validation, reliability, and discriminat
ory capacity.', Título do artigo
"STUDY DESIGN: Observational study of patients with scoliosis and matched controls. OBJECTIVES: To determine the validity a
nd reliability of the modified Scoliosis Research Society Outcomes Instrument (SRS-22) for use in the assessment of deformity
in adults. To demonstrate the discriminate validity of the SRS-22 in differentiating between affected and unaffected adults.
SUMMARY OF BACKGROUND DATA: Spinal deformity has an important impact on the general health status of adults. The magnitude of
this impact has been difficult to measure and reported variably in the literature. The development of disease-specific outcom
es tools permits improved sensitivity and specificity in measuring the patient's self-assessment of health status. An instrum
ent for measuring disease-specific health status in adults with scoliosis has not been validated. MATERIALS AND METHODS: Obse
rvational study comparing the health status of adults affected by scoliosis and unaffected controls, matched for age, gender,
and socioeconomic status. Pearson correlation analysis was used to determine the relationship of each domain within the SRS-2
2 and SF-36 with radiographic parameters including sagittal balance, coronal balance, and major curve correction. Discriminat
e validity of the modified SRS instrument was determined by a comparison of means between affected and unaffected cohorts. Th
e validation of the SRS-22 was determined by criterion validity, using correlation analysis with comparable domains of the SF
-36. The reliability of the SRS-22 was demonstrated using test-retest parity and Cronbach's alpha test for internal consisten
cy. RESULTS: One hundred eighty adults were included in the study, 146 with scoliosis and 34 without. Adults with scoliosis s
cored significantly lower than unaffected controls on every domain of the SRS-22 and the SF-36. The floor and ceiling effect
of the SRS-22 were less than observed in the SF-36 instrument. Pearson correlation analysis demonstrated a significant correla

```

Figure 18 - A demonstration of the content of the list formed from the two files.

With this sequence of steps so far, the result is a list such as the one seen in Fig. 18, containing 327 articles, which are not yet usable by the classifier and must still undergo text processing.

For greater precision in the *corpus* classification stage, one further keyword is introduced into the list: the *index* field. This field will make it possible for the data to be handled by the classifier algorithm with better access to the classification labels, which will help find the best hyperplane with the grid search technique for the classifier, as will be demonstrated later on.

First, however, the list with the four fields needs to be converted into a pandas dataframe. This is a rectangular data table representation containing an ordered collection of columns, each of which may hold a different type of value. What sets the pandas dataframe apart from a simple Python list is that it has an index for both rows and columns, which makes it easier both to view the content and to manipulate the data it contains, helping with the classification used by the algorithms. This process can be seen in Fig. 19 [106, 107].

Corpus (DataFrame)

	index	indice	classificacao	titulo	abstracti
0	0	138	1	Studies in the modified Scoliosis Research Soc...	STUDY DESIGN: Observational study of patients ...
1	1	192	0	Health related quality of life and patient rep...	OBJECTIVE: To evaluate health-related quality ...
2	2	197	0	Effectiveness and tolerability of the buprenor...	BACKGROUND: A new transdermal delivery system ...
3	3	198	1	Outcome of adenotonsillectomy for severe obstr...	OBJECTIVE: To study changes in sleep behavior ...
4	4	229	0	Gastric adenocarcinoma screening and preventio...	OBJECTIVE: To estimate the cost-effectiveness ...

Figure 19 - Corpus in pandas dataframe format with the index column added.

Once the *corpus* has been adjusted, the oversampling technique is used to balance the data. This technique is the most suitable for this type of situation, causing the minority sample to be balanced statistically without much loss of information.

Classificação 0: 265
 Classificação 1: 62
 Proporção: 4.27 : 1

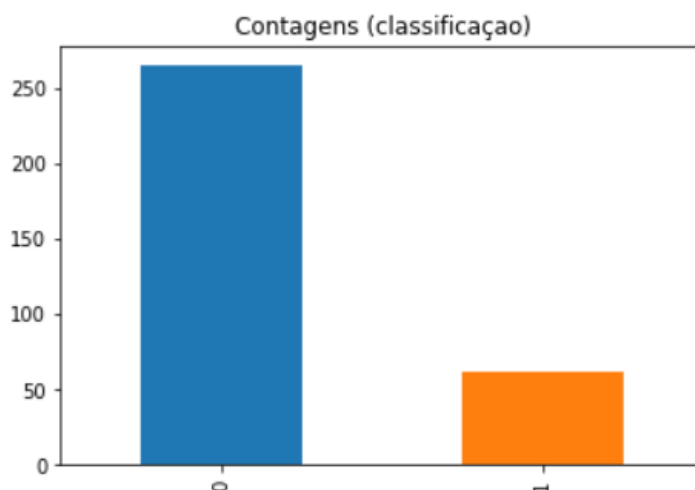


Figure 20 - Counts of the classifications before balancing.

In the chart of Fig. 20 it can be seen that the proportion of classified articles relative to the volume of the *corpus*, with a total of relevant classifications (1), is low, favoring the majority classification, not relevant (0).

Classificação 0: 265
Classificação 1: 265
Proporção: 1.0 : 1

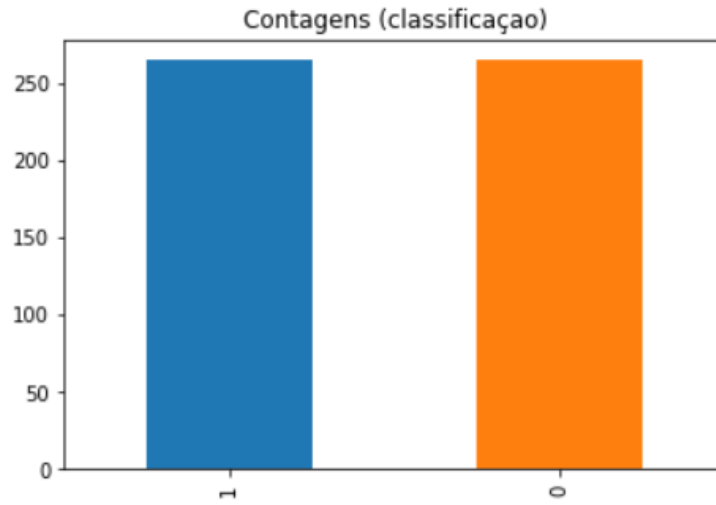


Figure 21 - Counts of the classifications after balancing.

With the use of the oversampling resampling technique, the *corpus* becomes balanced, synthesizing 203 additional articles for the positive classification and increasing the percentage of correct calls in the classification of the articles, with a total of 530 articles. This resampling technique was chosen because of the nature of the *corpus* on which this work is based, being a dataset of low dimensionality and with few “relevant” classifications. The oversampling method was the most suitable for restructuring the classification of the *corpus*, leaving the *corpus* at a proportion of 50% for both classes, as Fig. 21 shows, which helps with the supervised classification discussed further on.

4.2. Applying the probabilistic model together with the supervised classifier

The methodology used in this work has its foundation in supervised classification, using the SVM classifier. To obtain greater precision from this classifier, the LDA probabilistic model is used. Its purpose is to map all the topics that make up the articles contained in the *corpus*, so that the words in each document are captured chiefly by these imaginary topics (latent topics, those that go unnoticed in a reading). In order to obtain this resource, LDA uses representations based on clusters, which are induced across the whole set of articles, and its way of treating the articles differs from that of the SVM classifier [20, 48]. To carry out this procedure on the content of the *corpus*, preprocessing of the articles is required, as follows:

- Removal of punctuation and other kinds of unnecessary characters with the help of tokenization and regex-based NLP techniques.
- Removal of stopwords.

- Lemmatization: in this work this technique was used because it shows fewer losses in the sentences, or in the sense of the phrases, when terms are joined using the unigram and bigram technique.

After this procedure, the *corpus* produced by the screening is better suited to passing through the classification process proposed for supporting the SR.

Since the probabilistic method used in this dissertation is provided by the Gensim library, it creates two types of data used by the LDA statistical model, namely: the dictionary and the *corpus*, which in this case is represented as a BoW model. The dictionary holds a record of all the words contained in the *corpus*, assigning an identifier to each term, as can be seen in Fig. 22.

The BoW generated by LDA is a fitting of all the documents, having in its composition the matrix representation between articles and words. With the words represented by numeric identifiers and the frequency count occurring in the document, its use is for the purpose of searching for patterns in the term–document matrix; this can be seen in Fig. 23 [18, 64].

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In [27]: 1 print(id2word.id2token)

{0: 'actrn', 1: 'adelaide', 2: 'adherence', 3: 'adjusted', 4: 'advancement', 5: 'ambulatory', 6: 'among', 7: 'anzctr', 8: 'apnea', 9: 'appropriate', 10: 'a', 12: 'baseline', 13: 'bound', 14: 'care', 15: 'center', 16: 'centers', 17: 'change', 18: 'ci', 19: 'clinical', 20: 'comparable', 21: 'compare', 22: 'co', 23: 'conservative', 24: 'continuous_positive_airway_pressure', 25: 'controlled', 26: 'costs', 27: 'costs', 28: 'daytime_sleepiness', 29: 'decreased', 30: 'dem', 31: 'diagnosis', 32: 'difference', 33: 'differences', 34: 'disease', 35: 'due', 36: 'efficacy', 37: 'epworth_sleepiness_scale', 38: 'ess', 39: 'general', 40: 'group', 41: 'groups', 42: 'growing', 43: 'health', 44: 'high', 45: 'hospital', 46: 'ideally', 47: 'ideally', 48: 'identifier', 49: 'importance', 50: 'improvements', 51: 'increasing', 52: 'interest', 53: 'interventions', 54: 'involving', 55: 'june', 56: 'level', 57: 'life', 58: 'lower', 59: 'main', 60: 'management', 61: 'man', 62: 'mean', 63: 'mean', 64: 'mean', 65: 'mean', 66: 'measures', 67: 'medicine', 68: 'metropolitan', 69: 'model', 70: 'models', 71: 'modes', 72: 'month', 73: 'months', 74: 'noninferior', 75: 'objective', 76: 'obstructive', 77: 'org', 78: 'outcome', 79: 'outcomes', 80: 'patient', 81: 'patients', 82: 'physicians', 83: 'plans', 84: 'point', 85: 'practices', 86: 'primary', 87: 'quality', 88: 'randomized', 89: 'ranges', 90: 'regions', 91: 'relative', 92: 'relevance', 93: 'result', 94: 'resu', 95: 'rural', 96: 'satisfaction', 97: 'score', 98: 'scores', 99: 'secondary', 100: 'september', 101: 'services', 102: 'setting', 103: 'seventeen', 104: 'simplified', 105: 'sleep', 106: 'sleepiness', 107: 'south', 108: 'specialist', 109: 'specific', 110: 'splints', 111: 'study', 112: 'suggest', 113: 'tools', 114: 'training', 115: 'treated', 116: 'treatment', 117: 'trial', 118: 'trial_registration', 119: 'university', 120: 'using', 121: 'usual', 122: 'withdrawn', 123: 'within', 124: 'worse', 125: 'according', 126: 'age', 127: 'aged', 128: 'aim', 129: 'arthritis', 130: 'based', 131: 'better', 132: 'cance', 133: 'comparison', 134: 'completed', 135: 'conclusion', 136: 'conducted', 137: 'core', 138: 'data', 139: 'date', 140: 'declined', 141: 'depression', 142: 'diseases', 143: 'education', 144: 'eligible', 145: 'eortc', 146: 'eortc_q1q', 147: 'european_organization', 148: 'evaluating', 149: 'fatigue', 150: 'fema', 151: 'frequently', 152: 'function', 153: 'functioning', 154: 'generally', 155: 'generate', 156: 'german', 157: 'germany', 158: 'higher', 159: 'hyperlipide', 160: 'increased', 161: 'insomnia', 162: 'items', 163: 'latter', 164: 'lifetime', 165: 'maximum', 166: 'measured', 167: 'men', 168: 'methods', 169: 'mirored', 170: 'missing', 171: 'normative', 172: 'northern', 173: 'older', 174: 'pain', 175: 'participants', 176: 'persons', 177: 'population', 178: 'post', 179: 'prevalence', 180: 'procedures', 181: 'proportion', 182: 'provided', 183: 'published', 184: 'old', 185: 'ool', 186: 'questionnaire', 187: 'questi'}
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Figure 22 - A small example of the dictionary used by the LDA model.

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Doc1: 2
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```

in the articles, since they can distort the results; the bound is therefore set between 0.2% and 80%. In this way, the topics will be formed only by words with occurrences in this interval [39, 108].

With the *corpus* and the dictionary processed by LDA, the number of topics can be defined, where k is the representation of that number, denoted among the values $k = 10, 20, 50, 100, 150$ and 200 . In reality, however, there is no value of k that serves all topic-based events. This number is generally based on prior knowledge of the information base one has. In the case of a small collection of texts, this number of topics will obviously not be very large.

The value of k helps improve the representation of the terms in topics, its access being to the base of articles present in the *corpus*. For this, the value uses two hyperparameters, α and β , which help define whether the topics will be more general or more detailed [39].

$$\alpha = \frac{1}{\text{number}_{\text{topics}}} \quad (26)$$

To obtain this information from the topic model, it is necessary to pass as parameters the *corpus*, the dictionary, and the desired number of topics to the function representing LDA. These are processed through the iteration of each value of k , with the results stored in a variable created by the model, named by default *model*, or LDA model, containing its model for the *corpus* under analysis. This class is accessed through the Gensim library for later evaluation of the extracted topics. Fig. 24 shows a short excerpt of the algorithm representing the LDA model.

```

2 //Execução do LDA baseado na biblioteca Gensim
3
4 dictionary = gensim.corpora.Dictionary.load ('dictionary.dict')
5 corpus = gensim.corpora.MmCorpus ('corpus.mm')
6 Lda = gensim.models.ldamodel.LdaModel
7 ldamodel = Lda (corpus, num_topics=50, id2word = dictionary)
8
9 print (ldamodel.print_topics (num_topics=50, num_words=10))
0
1 ldamodel.save ('lda/lda50.model')
2

```

Figure 24 - Pseudocode of the LDA model. Note: the `num_topics` parameter is the same as the term k .

An interesting fact to note in the LDA model function in Python is the *passes* parameter (Fig. 25). This controls the frequencies, or cycles, with which the model's topics are grouped across the whole *corpus*, carrying out a review of each stage of the cycle that the

LDA class performs. This matter may be somewhat technical, but it is essential for controlling the quality of the model that will be delivered. It is important to set the number of passes so that the topic iterations are sufficient; the higher the number of passes, the longer the execution of the LDA class will take, on account of the reviewing.

According to [108, 109], in order to choose the most exact number of pass iterations it is advisable to enable the parameter that controls log perplexity (equation 24), setting it to a value of 1. Processing the model with this parameter set to 1 will produce log lines similar to the following:

```
2019-11-30 02:55:08,650 - gensim.models.MODELO_LDA - DEBUG - 50/530 documents
converged within 100 iterations
```

If *passes* = 25 is set, this line above will be observed 25 times for an exact number. If the number of times the line above appears is not equal to 25, then it may be concluded that the number in the *passes* variable is not ideal, and may be either more or less than 25 [109]. It is necessary to make sure that by the final cycles most of the documents have converged. If everything happens as described, the number returned by log perplexity will be the ideal one for taking part in the iterations of the LDA model. Another important factor to develop for a good topic model is the question of parallelism; this work was developed with processes of this technology.

The Gensim library makes available a technique by which the user can make use of the cores of the computer's processor, thereby giving greater speed to the process of building the LDA model. This step is marked by the parameter `workers = cpu_count - 1`, where `cpu_count` is a variable that will hold the number of CPUs. The `-1` is a way of preventing the user of the library from falling into the temptation of using all the processor cores [109]. Fig. 25 below shows the configuration of the LDA model used by the classification algorithm.

```
#modelos de LDA, Mallet, Coerência e Perplexidade
#criando modelo LDA (obs: o método LdaMultiCore é utilizado espeicilmente para fins de processadores multicore)

cpu_count = 8

#%time
MODELO_LDA = LdaMulticore(corpus          =corpus,      # corpus
                          id2word         =id2word,      #dicionário
                          random_state    =10,           #Faz o embaralhamento aleatório do corpus, quanto maior o numero de
                          num_topics       =10,           #O número de tópicos latentes solicitados a serem extraídos do corpus
                          passes          =25,           #Número de passes pelo corpus durante o treinamento, faz uma revisão
                          chunksize        =100,          #Número de documentos a serem usados em cada bloco de treinamento.
                          batch            =False,
                          alpha            ='asymmetric',#multicore
                          decay            =0.5,
                          offset           =64,           #Hiper-parâmetro que controla quanto desaceleraremos os primeiros
                          eta              =None,
                          eval_every       =1,           #A perplexidade do log é estimada a cada atualização. Definir isso
                          iterations       =100,          #Número máximo de iterações através do corpus ao inferir a distrib
                          gamma_threshold  =0.002,        #Alteração mínima no valor dos parâmetros gama para continuar iter
                          per_word_topics  =True,
                          workers          = cpu_count -1 )
```

Figure 25 - LDA in Python: the parameters that make up the LDA function.

In this work an LDA model was proposed with the value $k = 100$ and a number of *passes* = 25. The same configuration will later be used for the Mallet library. According to the authors of [110, 111], Mallet is a Java-based package for statistical natural language processing, document classification, clustering, topic modeling, information extraction, and other machine learning applications on text.

Mallet uses Latent Dirichlet Allocation (LDA); Pachinko Allocation (PAM), a model that captures arbitrary, nested, and possibly sparse correlations between topics using a directed acyclic graph, providing a flexible alternative to the recent work of Blei and Lafferty (2006), which captures correlations only between pairs of topics [112]; and Hierarchical Latent Dirichlet Allocation (hLDA) for topic modeling, in which topics are gathered into a hierarchy using the processes whose structure is learned from the data [113].

This library offers better topic quality than the pure LDA technique. Gensim makes available a wrapper for implementing LDA together with Mallet. The LDA model used in conjunction with the Mallet package is built with 100 different topics, where each topic is a combination of key terms and each key term contributes a certain weighting to the topic, as seen in section 3.5.2.

Uso do Mallet Modelo de Tópicos <pre> [[55, '0.233*qol' + 0.080*life' + 0.062*quality' + 0.038*study' + ' '0.025*significant' + 0.025*hr' + 0.021*social' + 0.019*version' + ' '0.017*whoqol' + 0.016*characteristics'), (50, '0.083*osa' + 0.062*apnea' + 0.062*obstructive' + 0.056*sleep' + ' '0.042*quality' + 0.039*score' + 0.031*children' + 0.027*life' + ' '0.027*symptom' + 0.025*fatigue'), (32, '0.051*questionnaire' + 0.052*age' + 0.036*high' + 0.034*score' + ' '0.030*psychological_distress' + 0.028*relationship' + 0.026*median' + ' '0.023*assessment' + 0.021*multiple_tender_points' + 0.019*subjects'), (73, '0.056*analysis' + 0.049*data' + 0.035*costs' + 0.028*cattle' + ' '0.028*cost' + 0.023*estimated' + 0.023*traditional' + 0.023*receiving' + ' '0.021*forms' + 0.021*metoclopramide'), (41, '0.061*pain' + 0.044*months' + 0.037*chronic' + 0.033*methods' + ' '0.031*life' + 0.031*active' + 0.030*hospital' + 0.028*stage' + ' '0.022*groups' + 0.022*mesh'), (28, '0.070*insomnia' + 0.041*sleep' + 0.041*treatment' + 0.026*cbt' + ' '0.022*group' + 0.019*hypnotic_drug' + 0.018*treated' + 0.016*outcomes' + ' '0.016*months' + 0.015*month'), (14, '0.077*screening' + 0.043*cost_effectiveness' + 0.040*cost' + ' '0.027*based' + 0.024*risk' + 0.022*costs' + 0.022*treatment' + ' '0.020*analysis' + 0.017*adjusted' + 0.017*strategy'), (85, '0.073*patients' + 0.063*life' + 0.055*quality' + 0.044*group' + ' '0.038*significant' + 0.033*functional' + 0.029*groups' + ' '0.026*cordectomy' + 0.024*statistically' + 0.022*study'), (72, '0.069*cancer' + 0.061*treatment' + 0.050*qol' + 0.032*ortic_gld' + ' '0.028*life' + 0.027*fatigue' + 0.024*scales' + 0.021*questionnaire' + ' '0.019*nausea_vomiting' + 0.018*survival'), </pre>	Uso do LDA Modelo de Tópicos <pre> [[98, '0.000*patients' + 0.000*degrees' + 0.000*implant_removal' + ' '0.000*significantly' + 0.000*time' + 0.000*assays' + 0.000*years' + ' '0.000*percent_predicted' + 0.000*instrumentation_none' + ' '0.000*similar_operation'), (99, '0.000*kccq' + 0.000*patients' + 0.000*heart_failure' + 0.000*measures' + ' '0.000*bnp' + 0.000*osa' + 0.000*changes' + 0.000*baseline' + ' '0.000*quality' + 0.000*status'), (96, '0.018*sleep' + 0.018*symptoms' + 0.016*one' + 0.016*gerd' + ' '0.015*vonoprazan' + 0.015*patients' + 0.014*associated' + ' '0.013*questionnaire' + 0.013*psychological_distress' + 0.012*year'), (97, '0.000*multinorbidity' + 0.000*among' + 0.000*positive' + 0.000*hiv' + ' '0.000*art' + 0.000*epilepsy' + 0.000*camps' + 0.000*appropriate' + ' '0.000*searched' + 0.000*membership'), (93, '0.044*kccq' + 0.018*correlation' + 0.018*perceived' + 0.016*score' + ' '0.016*functional' + 0.014*status' + 0.012*physician' + 0.011*hcm' + ' '0.011*nyha_class' + 0.010*correlations'), (94, '0.000*patients' + 0.000*depression' + 0.000*anxiety' + 0.000*life' + ' '0.000*effect' + 0.000*quality' + 0.000*study' + 0.000*significantly' + ' '0.000*epilepsy' + 0.000*improvements'), (95, '0.024*patients' + 0.016*epilepsy' + 0.014*scale' + ' '0.013*turkish_version' + 0.013*scores' + 0.012*life' + 0.012*quality' + ' '0.011*mini_qol' + 0.011*study' + 0.011*compared'), (92, '0.000*penital' + 0.000*skills' + 0.000*coping' + 0.000*screening' + ' '0.000*stress' + 0.000*hsv' + 0.000*treatment' + 0.000*quality' + ' '0.000*management' + 0.000*episodes'), (91, '0.000*patients' + 0.000*hen' + 0.000*quality' + 0.000*one' + ' '0.000*study' + 0.000*zygomatc_implants' + 0.000*procedure' + ' '0.000*suffer' + 0.000*inflamed' + 0.000*four'), </pre>
---	---

Figure 26 - Topics in the LDA model and the model being used with the Mallet library.

Fig. 26 may be interpreted as follows. Topic 72 of the classification carried out with the Mallet library will be analyzed. It can be described by the 10 main keywords that contribute to this topic, which are ‘cancer’, ‘treatment’, and so on, and the weight of ‘cancer’ in topic 72 is 0.069. The weights reflect the importance of a keyword to the chosen topic, offering a simple way of examining the theme that the topic is dealing with. In this case, for topic 72, it can be observed that the theme is cancer treatment and its consequences.

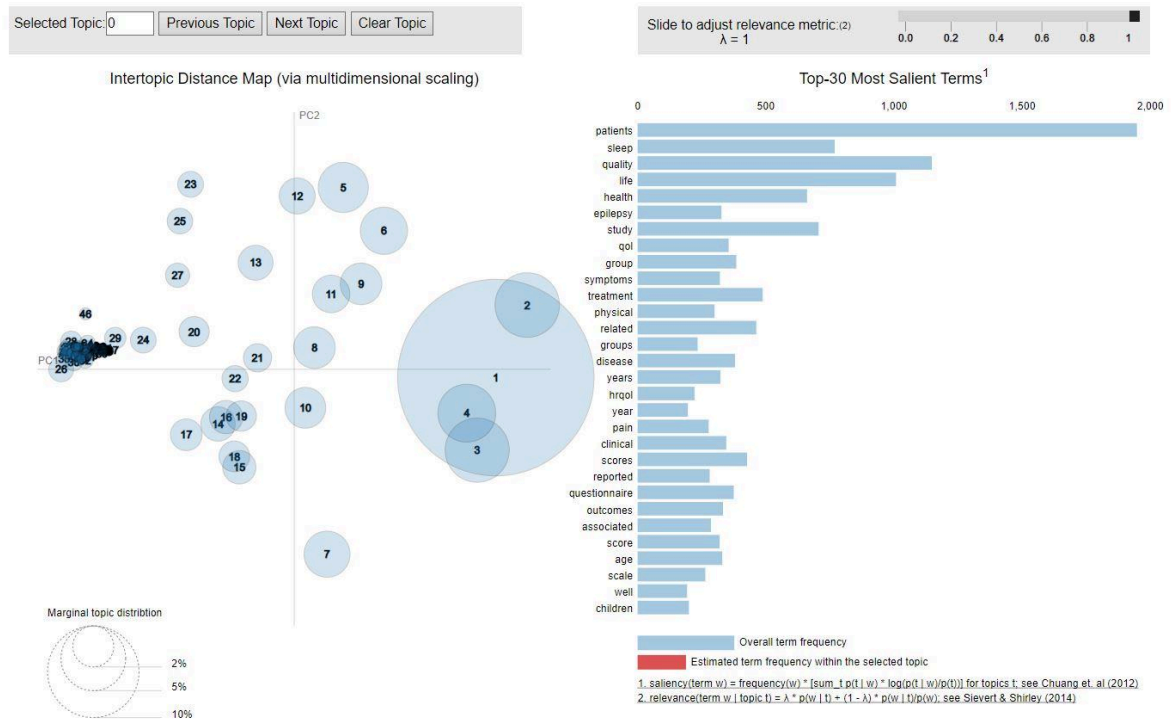


Figure 27 - Another way of viewing the main topics using the Python package pyLDAvis.

For a better visualization than that of Fig. 26, the pyLDAvis package, found in the Gensim library, is used. It offers a dynamic way of analyzing the terms that may be shared by different topics.

Fig. 27 shows that the bubbles in the chart on the left-hand side represent the topics, where their importance is measured by their size. When the topic model has large bubbles that do not overlap one another and are spread across the chart, it may be considered a good model. When k is very large, there will be overlapping of topics in small bubbles at a particular position in the chart, causing granularity in the subtopics that make up the topics [114].

So that no time is lost in manually searching for a suitable number of topics, it is often necessary to use calculations by means of coherence. This measure is taken through the ideal number of k topics, with $n = 1, 2, 3, \dots, k$ being analyzed for this question, and choosing the one that offers the highest coherence value (see section 3.5.2).

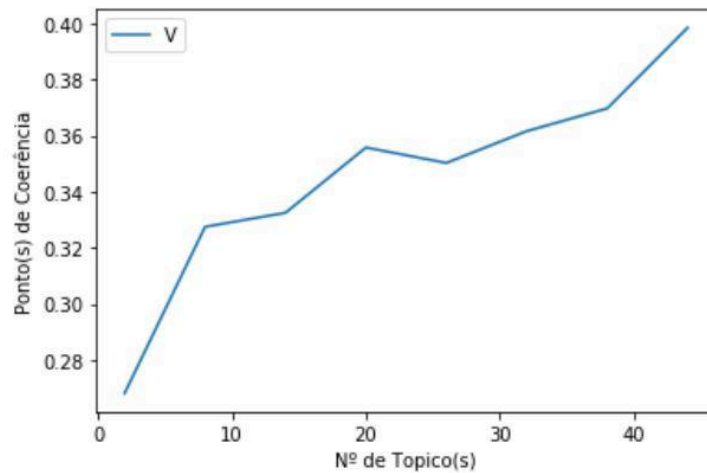


Figure 28 - Coherence points and coherence value.

The idea behind the chart in Fig. 28 is to give the researcher the number closest to the ideal number of topics, by building many LDA models with different values of k and choosing the one that gives the highest coherence value. Choosing a value of k that marks the end of the rapid growth in topic coherence generally offers more meaningful and interpretable topics. Choosing an even higher value can sometimes give more granular subtopics [108, 109].

If the topic contains the same keywords repeated across several neighboring topics, it is probably a sign that k is too large.

Nº Topicos = 2	tem um valor de coerência de 27.0 %
Nº Topicos = 8	tem um valor de coerência de 33.0 %
Nº Topicos = 14	tem um valor de coerência de 33.0 %
Nº Topicos = 20	tem um valor de coerência de 36.0 %
Nº Topicos = 26	tem um valor de coerência de 35.0 %
Nº Topicos = 32	tem um valor de coerência de 36.0 %
Nº Topicos = 38	tem um valor de coerência de 37.0 %
Nº Topicos = 44	tem um valor de coerência de 40.0 %

Maior Coerencia...: 40.0 %

Figure 29 - Coherence scores.

The value of k chosen in this context was $k = 20$, where it can be seen that at position 20 the chart begins to flatten out (Fig. 28) and shortly afterwards grows exponentially. With a very large value of k , as mentioned earlier, keywords may be repeated across several topics. This process of choosing the best number of topics takes a certain amount of time to consider, because the function developed for this task computes coherence for each article to be examined by the LDA model. Fig. 29 presents the coherence for each topic and shows the highest coherence among them.

Comparações com os valores de Coerências

```

Perplexidade do modelo.: -14.74
*****

Pontos de Coerência(Mallet).: 0.46
Pontos da Coerência(LDA).: 0.53

```

Figure 30 - Perplexity and coherence metrics used for the LDA and Mallet models.

As mentioned earlier in subchapter 3.5.3, the lower the log perplexity the lower the error rate, whereas for coherence points the higher the better. This can be confirmed in Fig. 30, with the comparison between the LDA model and LDA used together with Mallet.

Documento mais representativo para cada tópico

Nº_Topic	Topic_Perc_Contrib	Palavra_Chave	Abstract (Doc)
0	0.0	0.76 groups, degrees, study, questionnaire, analysi...	[214, 11268, 1, Discriminative validity of the...
1	1.0	0.83 treatment, sleep, quality, insomnia, patients,...	[35, 1482, 1, Psychological treatment for inso...
2	2.0	0.76 patients, group, life, disease, scores, compar...	[13, 623, 1, Epilepsy in patients with primary...
3	3.0	0.60 screening, results, data, included, based, adj...	[204, 10880, 0, Cost-effectiveness of differen...
4	4.0	0.81 study, sf, adults, health, specific, srs, asse...	[0, 138, 1, Studies in the modified Scoliosis ...
5	5.0	0.78 patients, follow, years, clinical, year, repor...	[178, 9493, 1, Clinical and radiographic resul...
6	6.0	0.54 sleep, quality, study, reported, symptoms, obs...	[304, 15893, 1, Assessment of the psychologica...
7	7.0	0.65 epilepsy, qol, seizure, variables, related, li...	[44, 1983, 1, Comorbidity of body dysmorphic d...
8	8.0	0.66 treatment, hiv, mg, months, study, improved, p...	[173, 9245, 1, Clinical outcomes using ClearPo...
9	9.0	0.46 qol, reported, symptoms, cancer, study, popula...	[279, 14718, 0, Italian National Survey of Dig...
10	10.0	0.62 patients, quality, life, surgery, complication...	[86, 3865, 1, Complications after surgery for ...
11	11.0	0.70 children, health, life, quality, study, relate...	[134, 6874, 1, Trajectories of health-related ...
12	12.0	0.76 score, kccq, status, health, patients, functio...	[171, 9190, 1, Correlations between physician-...
13	13.0	0.53 care, patients, patient, outcomes, disease, ma...	[307, 15957, 0, Practices of traditional beef ...
14	14.0	0.63 health, women, men, life, quality, related, sy...	[72, 3292, 1, Onabotulinum toxin A Injections ...
15	15.0	0.67 patients, hrqol, life, quality, questionnaire,...	[206, 10969, 1, Development of a Persian Versi...
16	16.0	0.68 patients, depression, pain, anxiety, symptoms,...	[161, 8473, 1, Prevalence of neuropathic pain ...
17	17.0	0.62 physical, function, weeks, quality, emotional,...	[303, 15812, 1, Health-related Quality of Life...
18	18.0	0.70 patients, therapy, intervention, years, group,...	[99, 4623, 1, Effect of Trimetazidine Dihydroc...
19	19.0	0.64 age, questionnaire, score, relationship, life,...	[150, 7723, 1, The relationship between psycho...

Figure 31 - Most representative document for each topic.

Fig. 31 shows the percentage contribution of the topics (keywords) in each document analyzed. With the results of the probabilistic model complete, a new *corpus* is obtained containing new parameters. This *corpus* assembled by LDA is used by the supervised SVM classifier, which has in its classification structure the aim of performing polarity analysis into “relevant” or “not relevant” for the case of this dissertation.

The feature extraction process is then begun, through the training of the classifier; in this dissertation the TF-IDF statistical measure is used together with the SVM classifier. To obtain greater performance from the SVM technique, it was proposed to find the best hyperparameter for this classifier, making adjustments to improve its performance and its learning rate.

SVM can generally offer many hyperparameters, and finding the right model for classification can make the task very arduous; for this task, however, the technique called grid search was used — a searcher for the best hyperparameter solution that will evaluate all the combinations passed through a dictionary and will use cross-validation calculations to arrive at that result [38, 115].

In Fig. 32 the internal structure of the grid search can be seen, with its processing cycles.

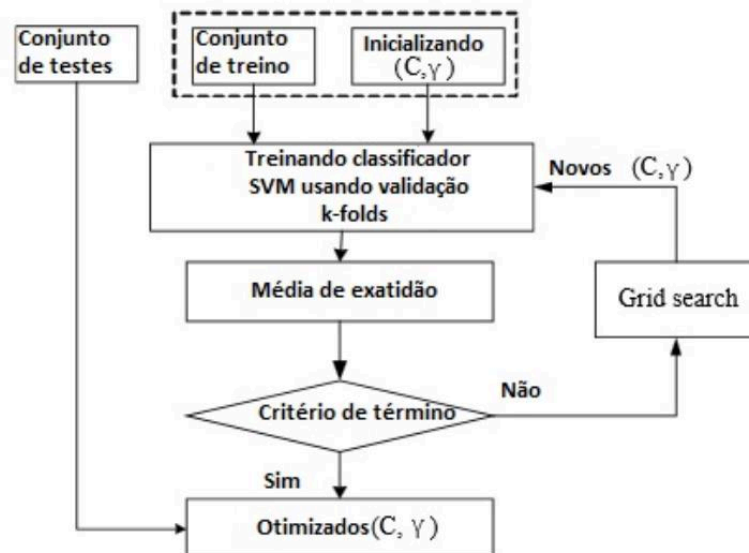


Figure 32 - Illustration of the cycle of a grid search in the quest for optimal parameters [115].

For this task it was necessary to divide the *corpus* in a proportion of 70% of the articles for training and 30% for testing, using the SVC() (support vector classification) function model from the scikit-learn library. This function is used in its default form because the dataset used in this dissertation is relatively small; in this case it will not be necessary to use more specific classes such as LinearSVC(), also belonging to the scikit-learn library, which performs better with large datasets [38, 117]. Without going through any adjustment in search of the best hyperparameters, it can be seen that the classifier has a proportion favorable to the “not relevant” class.

	precision	recall	f1-score	support
0	0.51	0.77	0.61	56
1	0.41	0.18	0.25	50
accuracy			0.49	106
macro avg	0.46	0.47	0.43	106
weighted avg	0.46	0.49	0.44	106

Figure 33 - Classes without any kind of hyperparameter adjustment.

As can be seen in Fig. 33, the classes classified by SVC, without any kind of hyperparameter adjustment, show in their form that the *corpus*, however much it has been balanced by the oversampling process, still has its differences as regards the degree of polarization.

In Fig. 34, on using the grid search method, the hyperparameter was adjusted for better precision in the “relevant” classification, 1 at 100%. Note that these results are based on the state of the *corpus* balanced with processing by the LDA statistical method.

	precision	recall	f1-score	support
0	0.95	1.00	0.97	56
1	1.00	0.94	0.97	50
accuracy			0.97	106
macro avg	0.97	0.97	0.97	106
weighted avg	0.97	0.97	0.97	106

Figure 34 - Classes with hyperparameter adjustment.

Best hyperparameter adjustment found using the grid search method:

```
Melhor parâmetro após o ajuste...: {'C': 1, 'gamma': 1, 'kernel': 'linear'}
Melhor Modelo para o SVM...:
SVC(C=1, gamma=1, kernel='linear')
```

Figure 35 - Adjusted hyperparameters with linear kernel.

where the parameters shown in Fig. 35 may be understood as:

C: regularization of the hyperparameter, with a value equal to 1.

Gamma: adjusting for linear hyperplanes; value 1, equal to linear.

Kernel: linear hyperplane, separating the “relevant” from the “not relevant.”

After the grid search process was complete, the best hyperparameter adjustment is taken to be $C = 1$ with kernel: linear, which offers an improvement in supervised classification, as will be demonstrated shortly after the results with the adjustment.

4.3. Feature extraction based on the k-fold technique

To evaluate the classifier model used, the cross-validation technique is employed, which performs the analysis from a dataset with the aim of estimating the precision of the classifier. This technique tends to add data to the part that will be used to train the classifier. For the development of cross-validation in this dissertation, the k-fold technique is used with the scikit-learn library.

The k-folds technique has at its basis the concept of dividing the set of information into smaller, mutually exclusive subsets, so that they can be used in estimating the parameters of the classification model (training data), leaving the rest of the data from the subsets (validation or test data) responsible for validating the model [38, 115].

The division in the cross-validation technique is governed by the following condition: the *corpus* to be analyzed in this example is denoted d to present the *corpus* as a single document, and it is divided into K parts, $(d_1, d_2, d_3, \dots, d_k)$. This choice is adopted by the developer of the classifier, but according to the author of [38] the standard is to divide it into 10 parts of similar size m_k , where $\sum_{k=1}^K m_k = n$, with K being the number of iterations of each validation sample, given by d_k , with $k = 1, 2, 3, 4, \dots, K$.

After this training sample of the predictor, the set of the other $K - 1$ parts is analyzed, remaining as follows: $d_{-k} = \{d_1, d_2, d_k - 1, d_k + 1, \dots, d_k\}$. At the end of the cross-validation process with the K cycles, the data are used both in the training part and in the validation part. This type of cross-validation, used in this dissertation, is called K -folds, and is exemplified by the following mathematical formula:

$$kfK = \frac{1}{K} \sum_{k=1}^K \frac{1}{m_k} \sum_{i=1}^{m_k} L(y_{ik}, \hat{f}_{(-k)}(x_{ik})) \quad (27)$$

where the predictor $\hat{f}_{(-k)}(x)$ is built for the training samples d_{-k} and is evaluated on the observations of the test sample d_k for every $k = 1, 2, 3, 4, \dots, K$ [115].

Cross-validation with the k -folds method performs a division of the data through the value K (the number of parts) between the training and test samples. In carrying out this task of dividing the documents in this validation technique, there is a tendency for the size of the samples m_k to decrease as the value of K is increased; a very high value results in a high computational cost, which implies small test samples and thereby increases the variance. According to scholars of the subject, the ideal value of K would lie between 2 and 10 [38, 15, 118].

In this dissertation the value $K = 4$ will be used, with the option of random number generation with 2 attempts. This randomness is the option of shuffling the content of the *corpus* so that it can be divided by the k-folds technique, this value giving better performance

for the type of *corpus* offered for this work. The reason is that these data are not very voluminous, so there is no need to slice the *corpus* into small units; on another occasion, however, the need to implement the k-folds cross-validation technique with a value of $K = 10$ would not be a bad idea.

Fig. 36 exemplifies the k-folds model with a dataset being divided into K parts of size m_k and with a training set of $K - 1$, that is, $d_{-k} = \{d_1, d_2, d_k - 1, d_k + 1, \dots, d_k\}$.

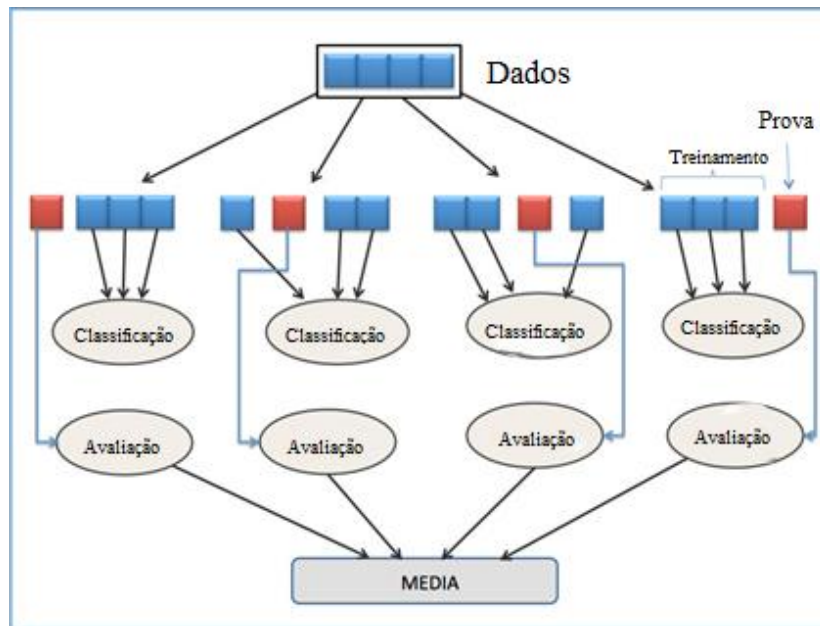


Figure 36 - Example of the implementation of the k-folds cross-validation technique.

With the value $K = 4$, the *corpus* was divided into 354 articles for training and 176 articles for testing, which are then processed for feature extraction with the TF-IDF statistical measure. Fig. 37 below presents a piece of the code for the part that extracts the features from the text files:

```
vectorizer = TfidfVectorizer(min_df=5,
                             max_df = 0.8,
                             sublinear_tf=True,
                             use_idf=True,
                             norm='l2',
                             encoding='latin-1',
                             ngram_range=(1, 2),
                             stop_words='english')

train_corpus_tfidf = vectorizer.fit_transform(X_train)
test_corpus_tfidf = vectorizer.transform(X_test)
```

Figure 37 - Demonstration of feature extraction from the articles with the TF-IDF statistical measure.

This resource creates the TF-IDF matrix, which computes the word count values, creates a model based on the TF-IDF measures, and helps convert text data into numeric

format, where the vector value it generates is the product of those two terms. TfidfVectorizer provides a vector named *vectorizer* so that it can be the feature model, the `fit_transform` function is called, a transformer for fitting the data instances. This function is called by the *vectorizer* vector, which takes as a parameter the set of articles for training and testing already divided by the k-folds method, these being iterable lists of strings in which each string represents the contents of the documents contained in the *corpus*, the content of these strings having a number of documents for each list. A short excerpt of the algorithm that forms the classifier is shown in Fig. 38 [38, 119, 76].

```
train_corpus_tf_idf = vectorizer.fit_transform(X_train)
test_corpus_tf_idf  = vectorizer.transform(X_test)
```

Figure 38 - The function is called by the *vectorizer* vector, which is the model coming from the TfidfVectorizer function; this function takes X_train and X_test as parameters.

The results generated by this `fit_transform` function are sparse matrices containing the string of numbers of each document of the iterable list, being the word count weighted by the TF-IDF statistical measure and the weights of the words per document of fixed length, for each document in the training and test sets. This may also be called the term–document matrix [76].

After the texts contained in the *corpus* have been processed into numeric format, the SVM classifier can be trained, with its hyperparameter already adjusted by the grid search technique. This classifier uses two parameters of the term–document matrix, these being training and test together with the polarity classification labels originating from the *corpus*. The content of these labels, as mentioned, forms part of the manual classification carried out by the human reviewers, which binds each article set aside for training and for testing. Fig. 39 shows how the labels are used in the form of vectors.

```
y_train, y_test = rotulos[train_index], rotulos[test_index]
```

Figure 39 - Extraction of the polarity labels of the classification.

The training of the SVM classifier is carried out together with the k-fold technique, and so are the TF-IDF weightings, not just the classifier. This can be explained by the fact that the cross-validation method had to be implemented inside a loop, where at each validation iteration both the TF-IDF part and the SVM part are processed.

At each iteration, the loop that makes up the cross-validation creates clones of the classifier, trains those clones on the training part, and makes predictions on the test part,

counting the number of correct predictions and delivering their proportion through the confusion matrix. This way of developing cross-validation is effective for gaining greater control over the data to be classified [38]. Fig. 40 shows an excerpt of the implementation of the k-folds technique in structured Portuguese (Portugol).

```

Para treino, teste ate K-folds(corpus, rotulos) faça
  Para cont ate treino faça
    x_traino <- corpus[cont]
  fim para

  Para cont ate teste faça
    x_teste <- corpus[cont]
  fim para

  y_traino <- rotulos[trano]
  y_teste <- rotulos[teste]

  vetorTFIDF <- TF-IDF()

  traino_corpus <- vetorTFIDF.tranformador(x_traino)
  teste_corpus <- vetorTFIDF.tranformador(x_teste)

  modeloClass <- classificadorSVC()

  modeloClass.tranformador(traino_corpus,y_traino)

  resultado <- modeloClass.predição(teste_corpus)
  Matriz_confusão_class <- Matriz_confusão_class + matrizConfusao(y_teste,resultado,rotulos[0,1])
fim Para

```

Figure 40 - Algorithm of the k-folds technique, written in structured algorithmic notation (Portugol). Demonstration of the interactions of cross-validation with the classifier and the TF-IDF method.

The k-folds cross-validation with $K = 4$ — that is, with 4 parts — made it possible to test all the files of the *corpus*, bearing in mind that, with the imbalance of this set of articles, the classification was practically polarized in equal proportion (see Fig. 21), leading to a fairer classification when obtaining the judgments for machine learning geared to the SR.

For a more detailed understanding of the algorithm proposed for supporting the SR, Fig. 41 shows in greater detail the whole development of the algorithm through flowchart modeling.

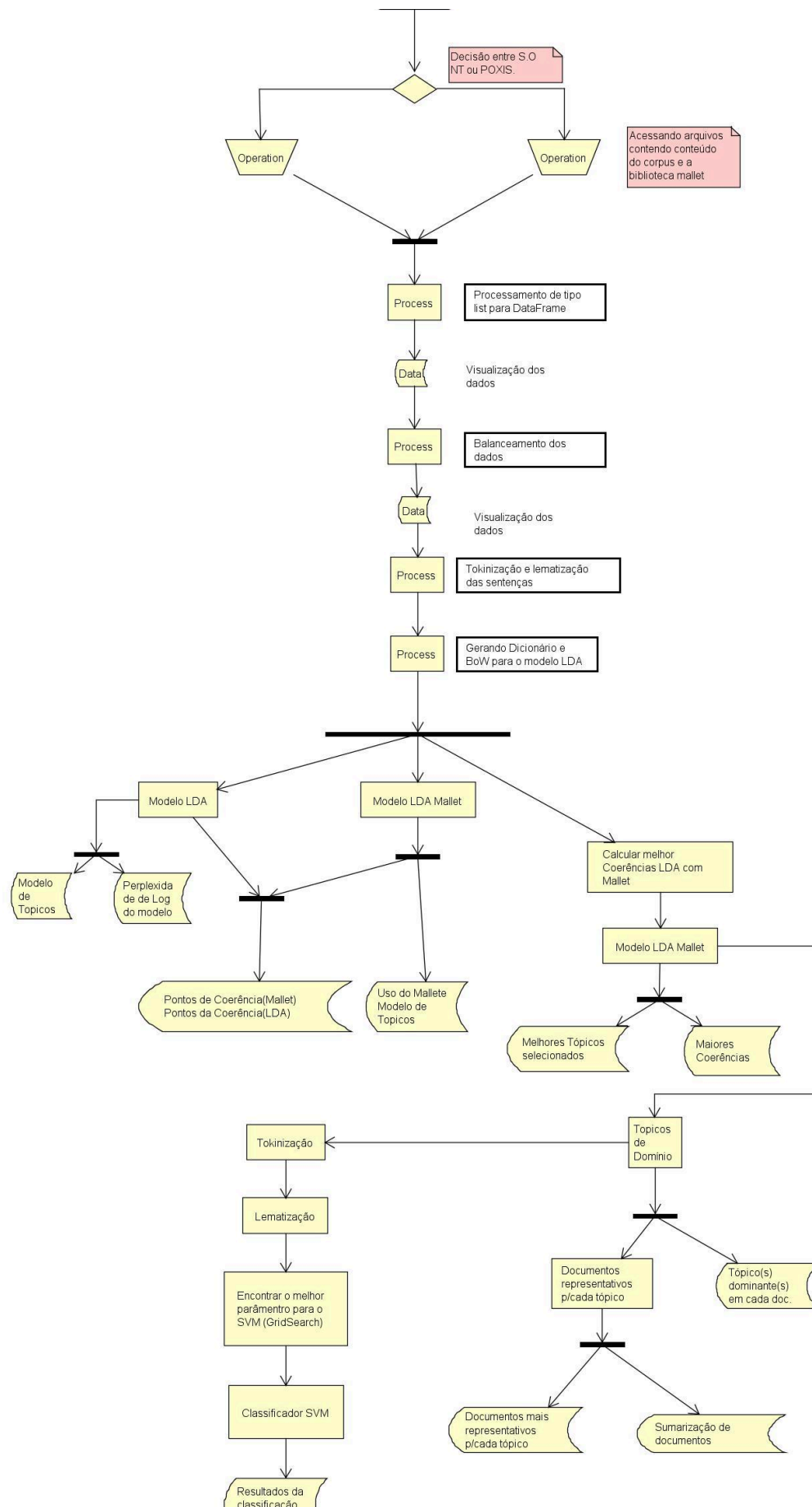


Figure 41 - Flowchart diagram representing each stage of the classifier algorithm.

4.4. Discussion of the results

With a total of 530 articles balanced and polarized as “relevant” and “not relevant,” the number of articles predicted correctly and the number of articles classified incorrectly by the classifier are presented, and alongside them a comparison can be made between the balanced and the unbalanced *corpus* with their classifications.

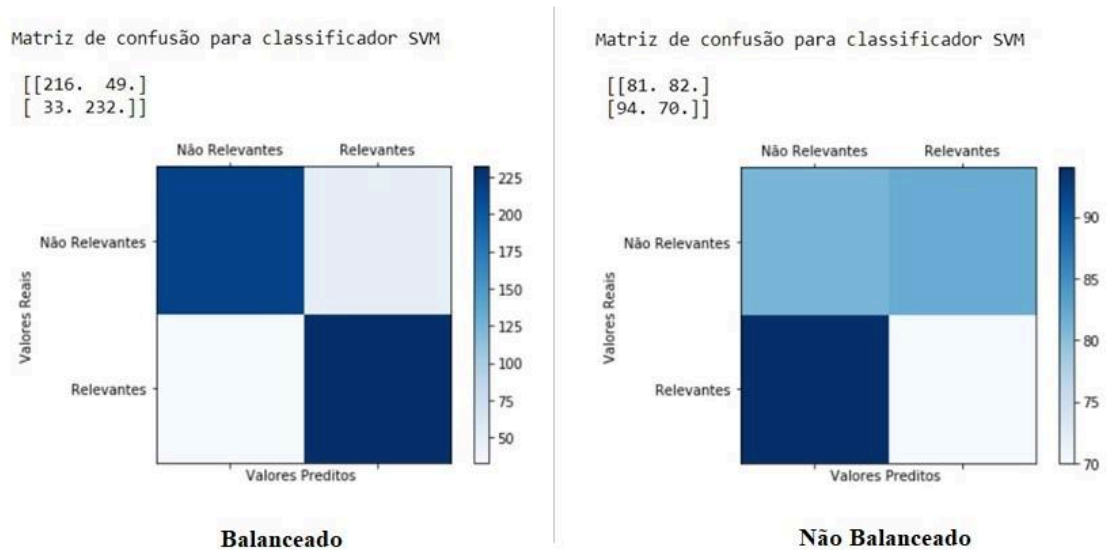


Figure 42 - Confusion matrix with the balanced and unbalanced corpus and their metrics.

As noted in subchapter 3.5.3, the confusion matrix exposes a great deal of information that needs to be analyzed in more detail. Fig. 42 shows, on the left-hand side, the confusion matrix with the classified and balanced *corpus*, and on the right-hand side the confusion matrix with the classification of the unbalanced *corpus*. Each row of the matrix represents an actual class, while each column represents a predicted class.

The first row of the balanced confusion matrix considers the “not relevant” articles: 216 of them were correctly classified as “not relevant,” which is called a true negative, TN. In the second quadrant, on the same row as the first quadrant, 49 articles were wrongly classified as “relevant,” which is a false positive, FP. In the second row, that is in the third quadrant, the “relevant” articles are considered: 33 articles were wrongly classified as “not relevant,” called a false negative, FN, while 232 articles were classified as “relevant,” which is called a true positive, TP.

One observation to be noted for a perfect classifier is that it would have only true positives TP and true negatives TN, so that the confusion matrix would have non-zero values only on its main diagonal (upper left to lower right). As can be seen, with this *corpus*

used with the hyperparameter adjustment and balanced, an almost perfect classification was obtained [38].

As observed in the matrix with the imbalanced *corpus* in Fig. 42, at a ratio of 4.27 “not relevant” articles to 1 “relevant” article, it can be seen that the concentration of articles as FN, false negative, in the imbalanced *corpus* is very large, with 94 articles wrongly classified as “not relevant.”

Following the same steps as the example of the results for the balanced *corpus*, it is notable that balancing the information contained in the dataset is important for obtaining answers with greater precision, helping both with classification and with the organization of the information.

It may be deduced that information, when it has a more balanced classification, yields as a result greater precision and confidence when using the classifier on the data provided. This can be seen in Table 4, which compares the metrics between the two types of *corpus*.

Tables 4 and 5 below present, with better visualization, the comparison of the metrics of the SVM classifier with the balanced and the unbalanced dataset.

Comparison of SVM classification with both types of corpus

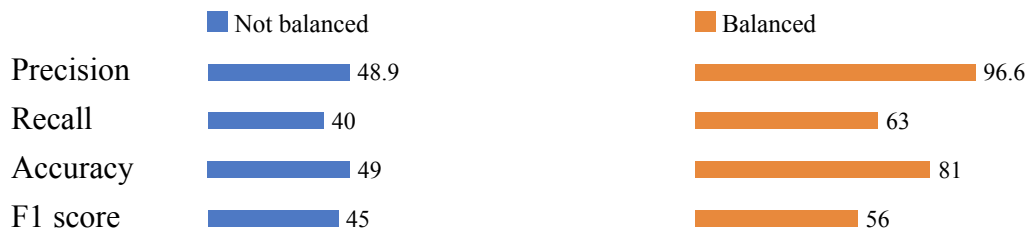


Table 4 - Comparison of the classification between both states of the corpus.

	Not balanced	Balanced
Precision	48.9	96.6
Recall	40	63
Accuracy	49	81
F1 score	45	56

Table 5 - Comparison of the classification between both states of the corpus, with their percentages.

As can be seen, there is a great difference between the two types of result for the two states of the *corpus*. The more balanced it is, the better the results will be, and the quantity of information will be taken into account, yielding a better proportion in the results.

To make the reviewers' work easier by sparing them from reading the whole abstract, which can be long, text summarization is used at this point in the classification of the articles. This helps the researcher gain a better advance sense of the content of the abstract, making it easier to know whether, even with the classification complete, that same article really would be what the researcher wants — bearing in mind that the supervised classifier proposed in this dissertation does not replace the human work of making a more complete and precise classification.

Fig. 43 shows the result of the summarization of the abstract of article Doc no. 16474; this field is the original index that the reviewers gave to each article out of a total of 530 contained in the balanced *corpus*. This summarization is based on the content analyzed statistically by the LDA model with Mallet, and can be carried out with the help of the Gensim library using the TextRank function.

Documento mais representativo para cada tópico - Abstract Sumarizado
Nº 1 Index: 296 Doc. Nº. 16474 Título: Quality of life among Brazilian children with epilepsy: validation of a parent proxy instrument (QVCE-50). Abstract: PURPOSE: To analyze the validity and reliability of an epilepsy specific health related quality of life (HRQL) instrument (QVCE-50), constructed for Brazilians. The QVCE-50 comprises the following domains: physical (9 items), psychological (18 items), social/familial (7 items), and cognitive/educational (16 items). QVCE-50 was analyzed for internal consistency, reliability, content and concurrent validity. RESULTS: Socio-demographic and clinical variables that differed in at least one domain were age, sex, time since diagnosis, epilepsy family history, and antiepileptic drug used. Internal consistency, analyzed by Cronbach's alpha, showed good results for total and domain scores: physical (0.68), psychological (0.86), socio-familial (0.70) and cognitive-educational (0.91).

Figure 43 - Summarization of the articles, covering the abstract of each document.

Fig. 44 presents what the content of the abstract of the same article was in the list before summarization.

```
[16474,
1,
'Quality of life among Brazilian children with epilepsy: validation of a parent proxy instrument (QVCE-50).',
"PURPOSE: To analyze the validity and reliability of an epilepsy specific health related quality of life (HRQL) instrument (QVCE-50), constructed for Brazilians. The QVCE-50 comprises the following domains: physical (9 items), psychological (18 items), social/familial (7 items), and cognitive/educational (16 items). Items were scored on a four point scale. Domains were equally weighted using percent scores. The questionnaire ends with a quality of life scale scored 0-10 and a space for free observations. METHODS: A total of 77 children with epilepsy were consecutively seen in a neuropaediatric ambulatory unit. Parents or caregivers responded to the QVCE-50 and a Portuguese version of ICIS (Impact of Childhood Illness Scale). QVCE-50 was analyzed for internal consistency, reliability, content and concurrent validity. Clinical and socio-demographic variables were also analyzed. RESULTS: Socio-demographic and clinical variables that differed in at least one domain were age, sex, time since diagnosis, epilepsy family history, and antiepileptic drug used. Internal consistency, analyzed by Cronbach's alpha, showed good results for total and domain scores: physical (0.68), psychological (0.86), socio-familial (0.70) and cognitive-educational (0.91). Total scores on the QVCE-50 and ICIS are inversely correlated (Pearson's r=-0.74, p<0.0001). The Intraclass Correlation Coefficients for test-retest reliability were acceptable: physical (0.51), psychological (0.62), socio-familial (0.66), cognitive-educational (0.85) and total (0.77), p<0.0001. CONCLUSIONS: QVCE-50 has good psychometric properties. It is a useful tool for analyzing HRQL in children with epilepsy for Portuguese speakers, especially Brazilians. Other properties should be further tested, such as responsiveness to drug and surgical treatment, capacity of distinguishing among seizure control
```

Figure 44 - Content of the article with index 16474 before summarization.

Together with the summarization, the recognition of the NEs contained in the abstract is also presented, as mentioned and explained in section 3.7. See Fig. 45.

Reconhecimento de entidade do Abstract.

PURPOSE: To analyze the validity and reliability of an epilepsy specific health related quality of life (HRQL PERSON) instrument (QVCE-50), constructed for Brazilians NORP . The QVCE-50 comprises the following domains: physical (9 CARDINAL items), psychological (18 CARDINAL items), social/familial (7 CARDINAL items), and cognitive/educational (16 CARDINAL items). Items were scored on a four CARDINAL point scale. Domains were equally weighted using percent scores. The questionnaire ends with a quality of life scale scored 0 CARDINAL -10 and a space for free observations. METHODS: A total of 77 CARDINAL children with epilepsy were consecutively seen in a neuropaediatric ambulatory ORG unit. Parents or caregivers responded to the QVCE-50 and a Portuguese NORP version of ICIS ORG (Impact of Childhood Illness Scale). QVCE-50 was analyzed for internal consistency, reliability, content and concurrent validity. Clinical and socio-demographic variables were also analyzed. RESULTS: Socio-

Figure 45 - Entity recognition in the summarization of the abstracts.

5. Conclusion

The motivation behind the development of this dissertation was to seek a solution that would make it possible to solve, or at least alleviate, the difficulty of performing the manual screening of articles used with the SR method. The existing estimate of the number of hours required to carry out the review is based on the title and abstract fields, and may last on average from days to several months, depending on the quantity of articles to be put through screening [13, 21].

To build the *corpus* used in this dissertation, a set of 17,132 articles was gathered from the PubMed search engine. Of that total, the reviewers managed to screen 327 articles, taking on average about 10 minutes to classify a single article as “relevant” or “not relevant.” To complete this task, the reviewers spent 54.5 hours classifying 327 articles, and this *corpus* is not balanced.

The classifier proposed for supporting the SR in article screening, built with a machine learning method together with text processing techniques such as text mining and NLP, is applied to the *corpus* with a ratio of 4.27:1 — that is, 265 articles classified as “not relevant” and 62 articles classified as “relevant” (see Fig. 20). To remedy this imbalance problem, the oversampling technique was used, allowing this same *corpus* to contain a greater proportion of “relevant” articles and thereby contributing to a more accurate classification.

With the balanced *corpus* of 530 articles processed by the classifier proposed for supporting the SR, a total of 232 articles were classified as relevant, with a precision of 96.6%, an accuracy of 81%, an F1 score of 56%, and a recall of 63%, as against the 216 articles classified as not relevant, taking on average 10 minutes to obtain the classification with the proposed, balanced *corpus* — a considerable margin of difference in time compared with the manual screening carried out by the researchers.

Out of curiosity as to whether classification would take longer with a high number of articles, this same dissertation used the whole proposed *corpus*, containing both the articles classified by the reviewers and the 16,804 articles that could not be classified, bearing in mind that what was measured was not classification accuracy but the time consumed. For this set of classified and unclassified articles, totaling 17,132, the same classifier algorithm was used, and it was faster than manual screening, taking on average 45 minutes to complete the classification of all the classified and unclassified articles, whereas for the same team of reviewers this time could reach 118 days, or 3 months and 28 days.

The machine learning classifier for semi-automating the classification process can help researchers in medicine as well as professionals in other fields of knowledge, such as biomedicine, law, and history — fields that demand a great deal of investigation — offering

greater speed and accuracy when researchers and reviewers obtain the articles they want, through classification using the title and abstract fields. This topic shows that machine learning in support of SRs helps people in their daily research work, semi-automating the process of searching for information in a faster and more practical way, and facilitating access to the literature through classifiers developed with machine learning methods.

5.1. Future work

The following topics are intended to be addressed in future work:

- Use the text summarization algorithm to reduce the content of the articles gathered, helping reviewers read the important parts of the whole article before the classification process and avoiding wasted time. The aim is thereby to obtain better manual screening before moving on to supervised classification.
- Automate the process of selecting articles in the PubMed and MEDLINE search engines with text summarization, as proposed in the first item of this section, but with the difference that this summarization process would be applied at the start of the selection process by the script developed in the present dissertation.
- In the classification process, use already summarized articles for training. These are expected to yield a better classification.
- Obtain the classification without going through supervised learning, making it possible to take advantage not only of the articles with defined classes but also of those that had no opportunity to be classified through supervised classification.

At the end of screening, researchers and reviewers will be able to worry less about the time needed to carry out the classification; using the summarization technique and unsupervised classification (not used in this dissertation) will save time and financial costs in the automatic screening process.

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Annex

Script used to collect the articles coming from PubMed.

```
import pandas as pd
import numpy as np
import pickle
from Bio import Entrez, Medline

file_loc = "keywordsbulk.xls"
df = pd.read_excel(file_loc, na_values=['NA'], usecols = "A")
df=df.dropna()
alpha = df['Alpha'].tolist()

df = pd.read_excel(file_loc, na_values=['NA'], usecols = "B")
df=df.dropna()
beta = df['Beta'].tolist()

df = pd.read_excel(file_loc, na_values=['NA'], usecols = "C")
df=df.dropna()
gamma = df['Gamma'].tolist()

tripleterms = []
for i in alpha:
    for j in beta:
        for k in gamma:
            aterm = i.lower() + ' ' + j.lower() + ' ' + k.lower()
            tripleterms.append(aterm)

doubleterms = []
for i in alpha:
    for j in beta:
        aterm = i.lower() + ' ' + j.lower()
        doubleterms.append(aterm)

allterms = tripleterms + doubleterms

Entrez.email = "goncalovaladao@gmail.com"
allpmids = []

for ttt in allterms:
    handle = Entrez.esearch(db='pubmed',
                            sort='relevance',
                            retmax='20',
                            term=ttt,
                            usehistory="y")

    pmids = Entrez.read(handle)['IdList']
    allpmids.extend(pmids)

allpmids = list(set(allpmids))

out_handle = open("auxiliary.txt", "w")
```

```

allpmidsaux = ','.join(allpmids)
fetch_handle = Entrez.efetch(db='pubmed',
                             rettype='medline',
                             retmode='text',
                             id=allpmidsaux)
data = fetch_handle.read()
fetch_handle.close()
out_handle.write(data)
out_handle.close()

with open("auxiliary.txt") as auxilhandle:
    therecords = Medline.parse(auxilhandle)
    recordslst = list(therecords)

auxilhandle.close()

with open('papers.dat','wb') as filename:
    pickle.dump(recordslst, filename)

```

List of keywords used in the script in order to access the articles coming from PubMed.

Symptoms / Manifestations	AND PRO term	AND QoL Terms
Cardiomyopathy	Patient reported outcomes	Quality of Life
Pericardial effusion	Patient reported outcome measures	Health related quality of life
Hepatomegaly	Observer reported outcomes	HRQOL
Cirrhosis	Observer reported outcome measures	Disease specific quality of life
Liver fibrosis	Caregiver reported outcomes	Symptom specific quality of life
Steatosis	Caregiver reported outcomes measures	
Feeding problems	Patient centered outcomes	
Feeding difficulties	Patient centred outcomes	
Gastrostomy	Proxy reported outcomes	
G-tube	proxy reported outcome measures	
Enteral feeding		
Diarrhea		
Vomiting		
Renal cysts		
Nephrotic syndrome		
Tubulopathy		
Osteopenia		
Kyphosis		
Scoliosis		
Joint contractures		
Hypotonia		
Psychomotor retardation		
Ataxia		
Hyporeflexia		
Stroke-like episodes		
Seizures		
Epilepsy		
Cerebellar hypoplasia		
Peripheral neuropathy		
Hypothyroidism		
Hypogonadism		
Thrombosis		
Development delay		
infections		
intellectual disability		
sleep disturbances		
strabismus		
retinitis pigmentosa		
nystagmus		
blindness		
behavioural problems		
behavioral problems		
behaviour		

Symptoms / Manifestations	AND PRO term	AND QoL Terms
behavior		
autism		
mood swings		
hyperinsulinemic		
hypoglicemia		
functional disability		
physical disability		

Link to the code and *corpus* used in the dissertation:

<https://github.com/michaelborges21/AUTOMATIC-LEARNING-TO-SUPPORT-SYSTEMATIC-LITERATURE-REVIEW>